

MACHINE LEARNING FOR PREDICTIVE DATA ANALYTICS

DOU LI

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Room2349, Science Building No.2



- 关于本课程
- AI历史回顾
- Machine Learning for Predictive Data Analytics

为什么要开这门课？

Undergraduate

**Undergraduate
Program**

Curriculum

Student Projects

Declaring ECE

ABET Accreditation

Princeton-Intel REU

Graduate

Courses

Undergraduate Program



为什么要开这门课？

普林斯顿大学的电气与计算机工程系（Department of Electrical and Computer Engineering, ECE）属于工程及应用科学研究学院（School of Engineering & Applied Science Research），该学院由以下六个系组成：

- Chemical and Biological Engineering 化学与生物工程
- Civil and Environmental Engineering 土木与环境工程
- Computer Science 计算机科学
- Electrical and Computer Engineering 电气与计算机工程
- Mechanical and Aerospace Engineering 机械与航天工程
- Operations Research and Financial Engineering 运筹学与金融工程

Rank	Program	Total score	Peer assessment	Employment rate at graduation	Employment rate three months after graduation	Average starting base salary plus sign-on bonus	Average GRE Quant of admitted FT students	Tuition	Cohort Size
1	Princeton University Master in Finance Princeton, NJ	100	4.0	100%	100%	\$155,571	168	\$107,540 (2 years)	29 FT
2	Baruch College, City University of New York Financial Engineering New York, NY	99	4.0	87%	95%	\$127,883	169.5	\$42,395 (non-resident) \$28,670 (resident)	29 FT, 1 PT
3	Carnegie Mellon University Computational Finance Pittsburgh, PA	98	4.4	78%	93%	\$117,775	169	\$86,820	98 FT
3	Columbia University Financial Engineering New York, NY	98	3.8	93%	100%	\$119,082	169.5	\$75,744	94 FT
5	University of California, Berkeley Financial Engineering Berkeley, CA	94	4.0	82%	88%	\$131,526	168.6	\$72,970	93 FT
6	New York University Mathematics in Finance New York, NY	93	3.8	60%	91%	\$115,000	169	\$84,000	46 FT, 3 PT
7	Cornell University MEng, FE concentration Ithaca, NY	92	3.6	76%	89%	\$127,773	168	\$84,825	78 FT
8	Columbia University Mathematics of Finance New York, NY	91	3.5	67%	94%	\$118,869	169.1	\$79,450	106 FT, 5 PT
9	NYU Tandon School of Engineering Financial Engineering Brooklyn, NY	87	3.3	73%	86%	\$96,961	169.2	\$70,000 (2 years)	157 FT
10	Massachusetts Institute of Technology Master of Finance	83	3.5	73%	94%	\$88,537	168	\$80,350 (12 months), \$111,275 (18 months)	116 FT

2025 Rankings

Rank	Program	Total Score	Peer Score	% Employed at Graduation	% Employed at 3 months	% Employed in the US	Compensation	Cohort Size	Acceptance Rate	Tuition
1	Princeton University 📍 Princeton, NJ 08544 Master in Finance ★★★★★ 7 reviews	100	3.7	95%	100%	90%	\$259,942	35 FT	4.3%	\$124,800
2	Baruch College 📍 New York, NY 10010 Master in Financial Engineering ★★★★★ 29 reviews	99	4.2	91%	100%	87%	\$200,207	23 FT, 4 PT	5.4%	\$42,395
3	Carnegie Mellon University 📍 New York, NY 10005 Pittsburgh, PA 15213 MS in Computational Finance ★★★★★ 45 reviews	93	4.2	89%	99%	97%	\$165,025	101 FT	16.8%	\$100,685
4	University of California, Berkeley 📍 Berkeley, CA 94720 Master of Financial Engineering ★★★★★ 37 reviews	88	3.6	79%	92%	83%	\$168,263	76 FT, 10 PT	18.0%	\$82,901
5	Massachusetts Institute of Technology 📍 Cambridge, MA 02139 Master of Finance ★★★★★ 26 reviews	86	3.0	76%	97%	66%	\$154,909	118 FT	8.7%	\$125,484
6	Columbia University 📍 New York, NY 10027 Masters in Financial Engineering ★★★★★ 11 reviews	85	3.6	37%	100%	56%	\$152,119	123 FT	10.5%	\$93,024
6	University of Chicago 📍 Chicago, IL 60637 MS in Financial Mathematics ★★★★★ 40 reviews	85	3.4	78%	95%	79%	\$134,285	124 FT, 6 PT	20.9%	\$96,826
8	Georgia Institute of Technology 📍 Atlanta, GA 30332 MS in Quantitative and Computational Finance ★★★★★ 9 reviews	80	3.0	93%	100%	89%	\$120,587	51 FT	32.6%	\$64,347
9	Columbia University 📍 New York, NY 10027 MA in Mathematics of Finance ★★★★★ 6 reviews	77	3.4	49%	75%	60%	\$129,249	102 FT, 7 PT	22.1%	\$98,930
9	Cornell University 📍 Ithaca, NY 14850 Master in Financial Engineering ★★★★★ 7 reviews	77	3.2	55%	84%	77%	\$140,500	65 FT	20.7%	\$102,570
9	NYU Tandon School of Engineering 📍 Brooklyn, NY 11201 MS in Financial Engineering ★★★★★ 30 reviews	77	3.0	67%	89%	64%	\$119,528	146 FT	28.1%	\$81,581



Rank	Program	Total Score	Peer Score	Employed at Graduation (%)	Employed at 3 months (%)	% Employed in the US	Graduates report rate	Compensation	Base salary	Salary report rate	Cohort Size	Acceptance Rate (%)	Yield Rate	Tuition
1	Baruch College 📍 New York, NY 10010 Master in Financial Engineering ★★★★★ 37 reviews	100	4.3	100	100	83	100	217.3K	178.8K	89	23	4.01	90.91	45.09K
2	Princeton University 📍 Princeton, NJ 08544 Master in Finance ★★★★★ 13 reviews	93	3.9	94	97	85	100	200.8K	160.2K	79	44	5.36	80	130.4K
3	Carnegie Mellon University 📍 New York, NY 10005 Pittsburgh, PA 15213 MS in Computational Finance ★★★★★ 55 reviews	91	4.3	72	100	100	100	161.6K	134.2K	85	108	17.22	59.67	104.6K
4	Columbia University 📍 New York, NY 10027 Masters in Financial Engineering ★★★★☆ 16 reviews	86	3.6	79	87	72	91	147.5K	138K	74	136	13.35	83.44	97.20K
5	Massachusetts Institute of Technology 📍 Cambridge, MA 02139 Master of Finance ★★★★☆ 27 reviews	84	3.2	78	98	50	100	167.8K	140.1K	86	126	8.33	70.39	129.4K
5	University of California, Berkeley 📍 Berkeley, CA 94720 Master of Financial Engineering ★★★★☆ 44 reviews	84	3.6	77	90	89	96	168.6K	154.3K	84	84	17.36	72.38	87.04K
7	University of Chicago 📍 Chicago, IL 60637 MS in Financial Mathematics ★★★★★ 52 reviews	79	3.6	73	91	84	98	140.2K	123.8K	100	124	22.01	36.42	100.6K
8	Georgia Institute of Technology 📍 Atlanta, GA 30332 MS in Quantitative and Computational Finance ★★★★☆ 12 reviews	76	2.9	94	96	100	84	120.7K	114.6K	100	107	30.18	44.39	66.13K
9	Columbia University 📍 New York, NY 10027 MA in Mathematics of Finance (MAFN) ★★★★★ 22 reviews	73	3.5	53	67	65	79	144.6K	122.6K	84	113	22.28	49.75	103.8K
9	North Carolina State University 📍 Raleigh, NC 27695 Master in Financial Mathematics ★★★★☆ 28 reviews	73	2.4	80	100	100	94	111.1K	105.3K	80	64	16.74	81.01	66.04K



为什么要开这门课？

<https://ece.princeton.edu/academics/undergraduate/curriculum>

10 concentrations of ECE:

- **Circuits and Systems**
- **Data and Information**
- **Computer Systems**
- **Security and Privacy**
- **Electronic Devices and Materials**
- **Biomedical Engineering**
- **Robotics and Cyber-physical Systems**
- **Quantum Information and Applied Physics**
- **Optics and Photonics**
- **Energy and the Environment**

ECE-Electrical and Computer Engineering

COS-Computer Science

ORF-Operations Research and Financial Engineering

ORF 309* Probability and Stochastic Systems (F & S) is required, plus two or three courses from this list.

Then two or three courses from the list.*

ECE 346 Intelligent Robotic Systems (S)

ECE 364*** Machine Learning for Predictive Data Analysis (F)

ECE 368 Intro to Wireless Communication Systems (S)

ECE 381 Networks: Friends, Money and Bytes (S)

ECE 382 Probabilistic Systems and Information Processing (S)

ECE/COS 432 Information Security (F & S)

ECE433/COS435** Introduction to Reinforcement Learning (S)

ECE434/COS 434**Machine Learning Theory (F)

ECE 435** Machine Learning and Pattern Recognition (F)

ECE 470 Principles of Blockchains (F)

→ ECE477 Smart Healthcare (S)

ECE 482 Digital Signal Processing (F)

ECE 483 Signal Processing and Optimization in Smart Grids – ONE-TIME COURSE (S '23)

ECE 486 Transmission and Compression of Information

→ ECE488 Fundamental Image Processing: From Mars to Hollywood with a Stop at the Hospital (F)

COS 302/ECE305** Mathematics for Numerical Computing & ML (F)

COS 324*** Introduction to Machine Learning (F)

COS 402** Artificial Intelligence

COS 424** Fundamentals of Machine Learning (S)

COS 429 Computer Vision (F)

ORF 350*** Analysis of Big Data (S)

ORF 363 Computing and Optimization for the Physical and Social Sciences (also COS 323) (F)

***Only one of the following machine learning courses may be applied to this concentration since they have overlapping content: ECE364, ORF350 and COS324.

COS 516 / ECE 516 <u>Automated Reasoning about Software</u>  Class #22350	L01	M W, 10:40 am – 12:00 pm, View Room	Seats Open: 29 Seats Enrolled: 11 Total Seats: 40	
ECE 115 <u>Introduction to Computing: Programming Autonomous Vehicles</u>  Class #21272	QCR L01	M W, 2:55 pm – 4:15 pm, View Room	Seats Open: 5 Seats Enrolled: 65 Total Seats: 70	
ECE 206 / COS 306 <u>Contemporary Logic Design</u>  Class #21248	SEL L01	T Th, 1:20 pm – 2:40 pm, View Room	Seats Open: 30 Seats Enrolled: 145 Total Seats: 175	
ECE 298 <u>Sophomore Independent Work</u>  Class #21253	S01	To be determined	Seats Open: 29 Seats Enrolled: 11 Total Seats: 40	
ECE 302 <u>Robotic and Autonomous Systems Lab</u>  Class #21259	L01	T Th, 12:15 pm – 1:05 pm, View Room	Seats Open: 1 Seats Enrolled: 89 Total Seats: 90	Closed
ECE 308 <u>Electronic and Photonic Devices</u>  Class #21254	SEL L01	M W, 10:40 am – 12:00 pm, View Room	Seats Open: 1 Seats Enrolled: 39 Total Seats: 40	
ECE 364 <u>Machine Learning for Predictive Data Analytics</u>  Class #21263	L01	M W, 2:55 pm – 4:15 pm, View Room	Seats Open: 2 Seats Enrolled: 70 Total Seats: 72	Closed
ECE 396 / COS 396 / QSE 320 <u>Introduction to Quantum Computing</u>  Class #21280	L01	M W, 10:40 am – 12:00 pm, View Room	Seats Open: 32 Seats Enrolled: 53 Total Seats: 85	

Machine Learning for Predictive Data Analytics

2026-2027 Fall

ECE 364

GRADING BASIS:

No Pass/D/Fail

INSTRUCTORS:

- [Niraj K. Jha](#)

LINKS:

- [Books](#)
- [Evaluations](#)

Description:

Machine learning for predictive data analytics; information-based learning; similarity-based learning; probability-based learning; error-based learning; deep learning; evaluation.

Sample Reading List:

- John D. Kelleher, Brian MacNamee, and Aoife D'Arcy, *Fund. of Machine Learning for Predictive Data Analytics, 2nd*

Reading/Writing Assignments:

Biweekly homeworks and project assignments.

Requirements/Grading:

Term Assessments:

- Exam(s) given during term - 25%
- Problem sets - 15%
- Programming assignments - 10%

Final Assessments:

- Final scheduled exam - 50%

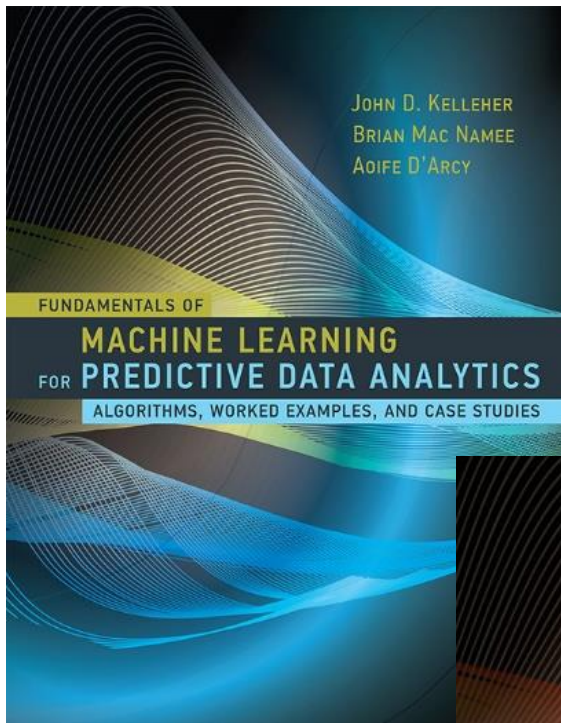
Other Requirements:

- Software and/or hardware use required
- Not Open to First Year Undergraduates.

Prerequisites and Restrictions:

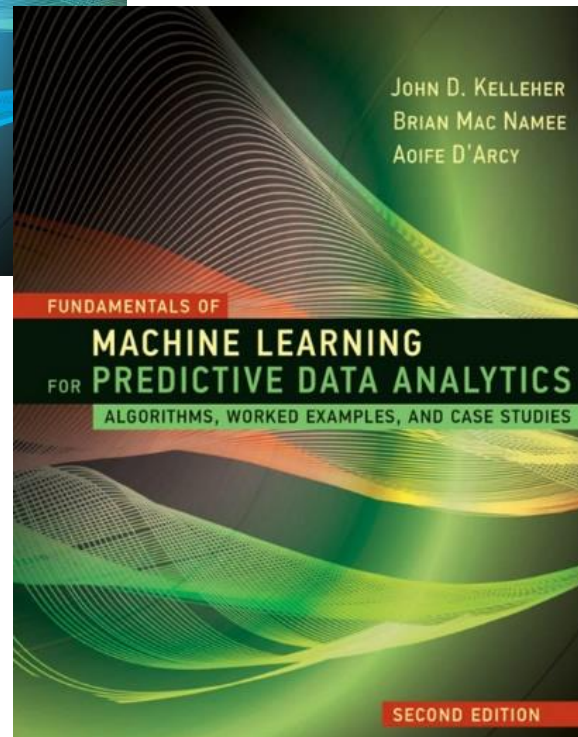
A strong background in differential and integral calculus, linear algebra, and probability is necessary.

教科书



2015

2020.10



The book is intended for use on machine learning, data mining, data analytics, or artificial intelligence modules on undergraduate and post-graduate computer science, natural and social science, engineering, and business courses.

II Predictive Data Analytics	115
4 Information-Based Learning	117
5 Similarity-Based Learning	181
6 Probability-Based Learning	243
7 Error-Based Learning	311
8 Deep Learning	381
9 Evaluation	533
III Beyond Prediction	595
10 Beyond Prediction: Unsupervised Learning	597
11 Beyond Prediction: Reinforcement Learning	637
IV Case Studies and Conclusions	683
12 Case Study: Customer Churn	685
13 Case Study: Galaxy Classification	703
14 The Art of Machine Learning for Prediction	729
V Appendices	743
A Descriptive Statistics and Data Visualization	745
B Introduction to Probability for Machine Learning	757
C Differentiation Techniques for Machine Learning	765
D Introduction to Linear Algebra	771
Bibliography	775
Index	787

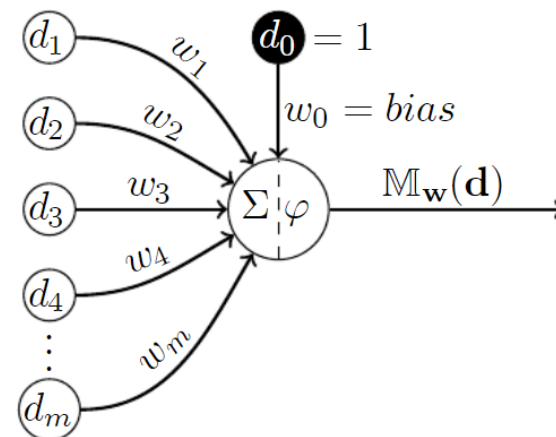


Figure 8.3
A schematic of an artificial neuron.

where $\mathbf{d}$ is a vector of $m + 1$ descriptive features (including the dummy $\mathbf{d}^{[0]}$ feature), $\mathbf{d}^{[0]}, \mathbf{d}^{[1]}, \dots, \mathbf{d}^{[m]}$; $\mathbf{w}$ is a vector of $m + 1$ weights (including the bias term) $\mathbf{w}^{[0]}, \mathbf{w}^{[1]}, \dots, \mathbf{w}^{[m]}$; and φ represents the activation function (threshold, tanh, logistic, or rectifier, etc.) that converts the result of the weighted sum to the output activation. Figure 8.3 presents a graphical representation of this same function. In this figure the arrows carry activations in the direction the arrow is pointing, the weight label on each arrow represents the weight that will be applied to the descriptive feature carried along the arrow, the Σ symbol represents the weighted sum of the inputs, and the φ symbol represents the threshold function being applied to the result of the weighted sum to convert it into an activation.



MY COURSE MATERIALS

SORT BY ▾

ECE 364 L01 Fall 2024

MACHINE LEARNING FOR PREDICTIV - HONGJIE WANG, JIAXIN XIAO, NIRAJ JHA, YIHAO LIANG

REQUIRED



FUNDAMENTALS OF MACHINE LEARNING FOR PREDICTIVE DATA ANALYTICS, SECOND EDITION ALGORITHMS, WORKED EXAMPLES, AND CASE STUDIES

KELLEHER, JOHN D.; MAC NAMEE, BRIAN; D'ARCY, AOIFE

ISBN13: 9780262044691

EDITION: 2ND

FORMAT: HARDCOVER

PUBLISHER: The MIT Press

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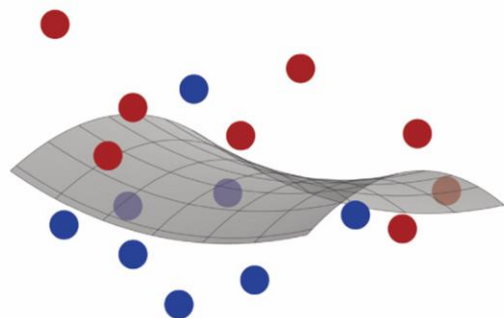
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ADD MORE COURSES >>>



Foundations of Machine Learning second edition



Mehryar Mohri,
Afshin Rostamizadeh,
and Ameet Talwalkar

Research Areas

The Courant Institute has a tradition of research which combines pure and applied mathematics, with a high level of interaction between different areas. Below we list some of the current areas of research. The choice of categories is somewhat arbitrary, as many faculty have interests that cut across boundaries, and the fields continue to evolve.

We give a very brief overview of the research in each area; more detailed information may be found on individual faculty webpages.

Computer Science

- Algorithms & Theory
- Computational Biology
- Formal Methods & Verification
- Graphics, Vision & User Interfaces
- Machine Learning
- Natural Language, Speech Processing, & Knowledge Representation
- Networks, Operating & Distributed Systems
- Scientific Computing

Mathematics

- Algebraic Geometry
- Analysis & PDE
- Computational & Mathematical Biology
- Dynamical Systems & Ergodic Theory
- Geometry
- Physical Applied Mathematics
- Probability Theory
- Scientific Computing

This book is a general introduction to machine learning that can serve as a reference book for researchers and a textbook for students. It covers fundamental modern topics in machine learning while providing the theoretical basis and conceptual tools needed for the discussion and justification of algorithms. It also describes several key aspects of the application of these algorithms.

We have aimed to present the most novel theoretical tools and concepts while giving concise proofs, even for relatively advanced results. In general, whenever possible, we have chosen to favor succinctness. Nevertheless, we discuss some crucial complex topics arising in machine learning and highlight several open research questions. Certain topics often merged with others or treated with insufficient attention are discussed separately here and with more emphasis: for example, a different chapter is reserved for multi-class classification, ranking, and regression.

Although we cover a very wide variety of important topics in machine learning, we have chosen to omit a few important ones, including graphical models and neural networks, both for the sake of brevity and because of the current lack of solid theoretical guarantees for some methods.

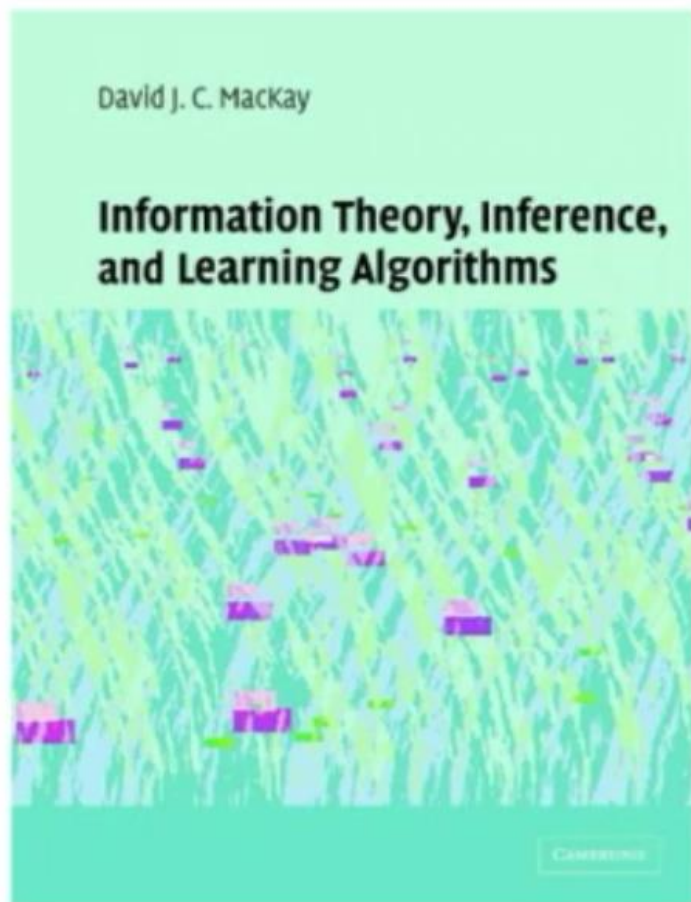
Mehryar Mohri (森)

Professor of Computer Science

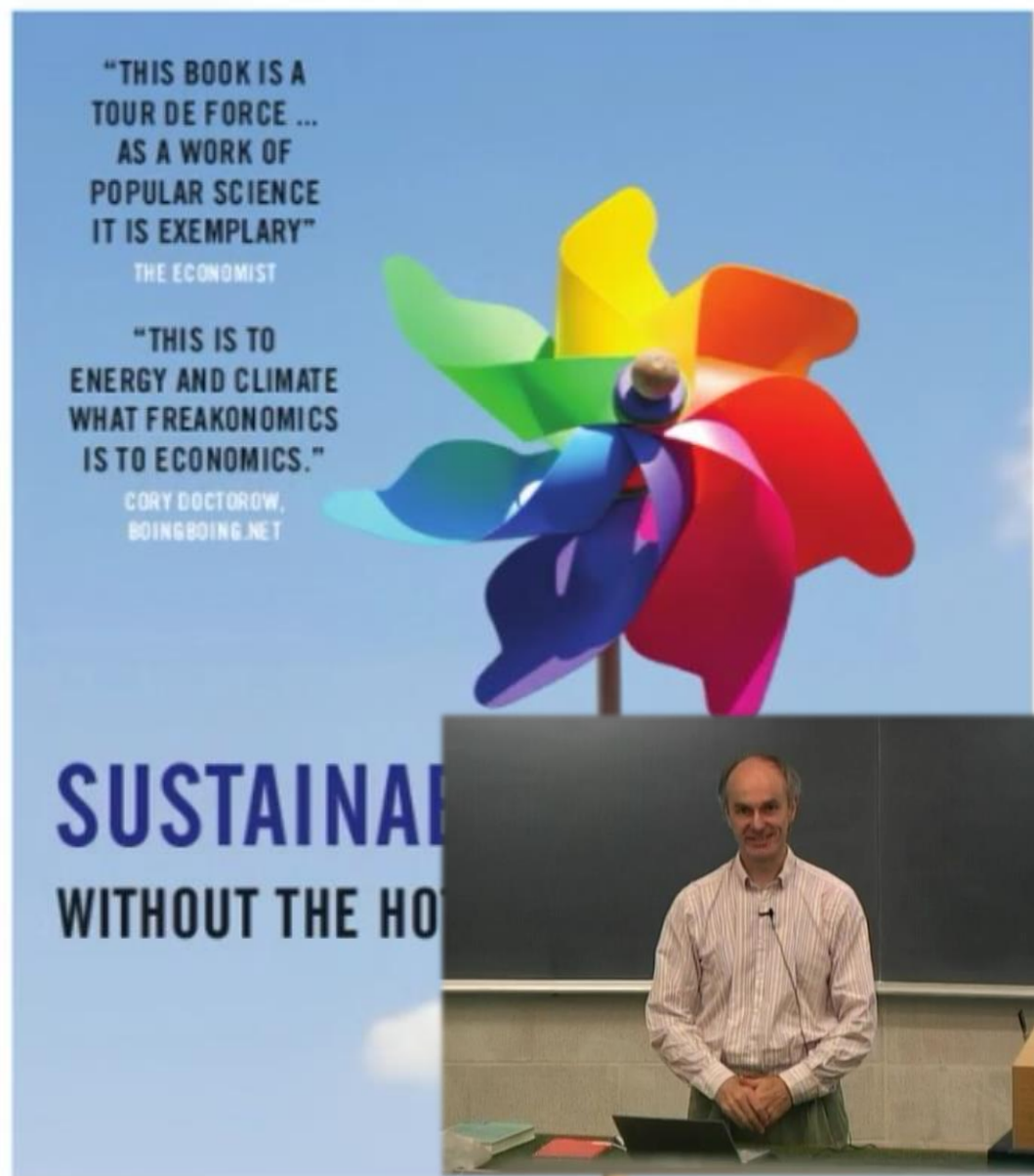
Courant Institute of Mathematics



Both available **free online**



www.inference.phy.cam.ac.uk/itila/

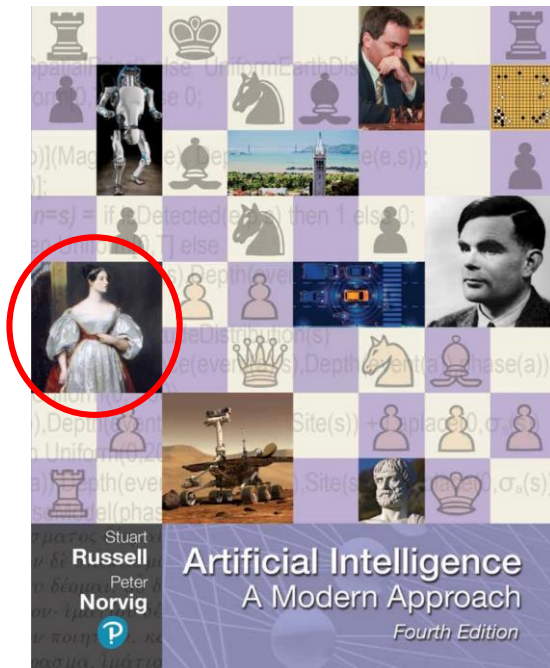


www.withouthotair.com



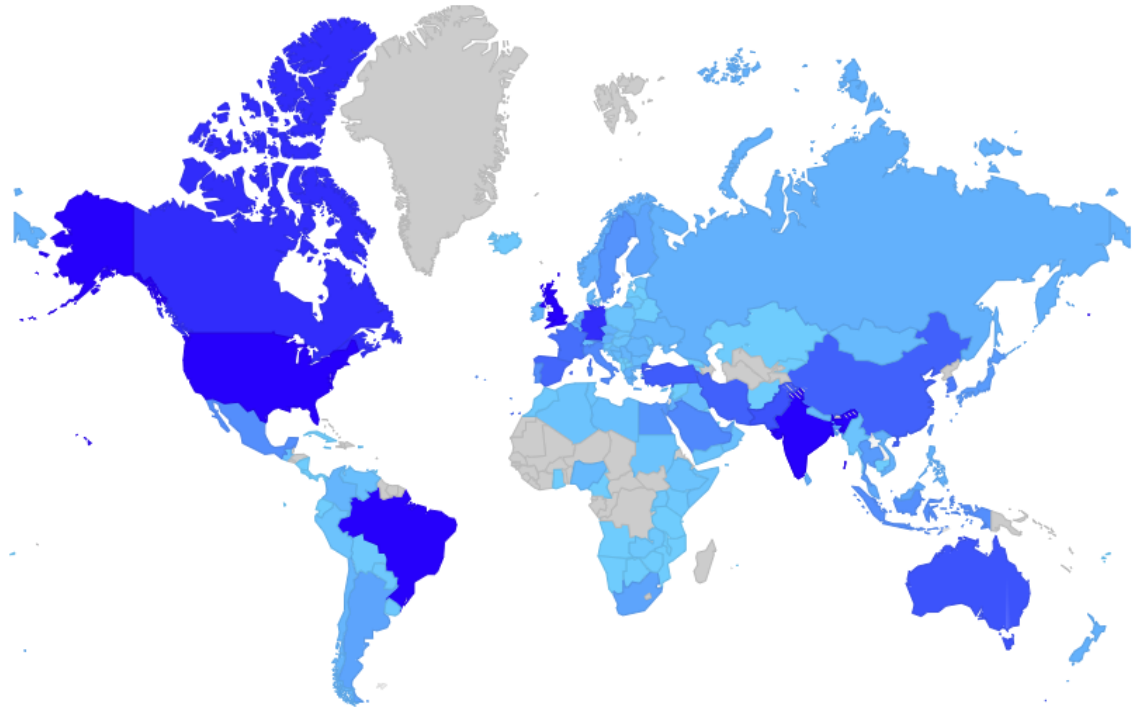
Stuart Russell

Professor of Computer Science, University of California, Berkeley



1542 Schools Worldwide That Have Adopted AIMA

Artificial Intelligence: A Modern Approach has been adopted for use by at least **1542 schools** in **134 countries or regions**. Please let peter@norvig.com know of any we missed. Hover over a country in the map or word cloud below to see the number of adoptions. Each @ after a school name links to a course home page.



Afghanistan Albania Angola Armenia Bahrain Barbados Belarus Botswana Cabo Verde Cambodia Cameroon Costa Rica Cuba Ecuador
 Fiji Georgia Ghana Kazakhstan Kosovo Kuwait Kyrgyzstan Latvia Luxembourg Macedonia Malawi Malta Mauritius Moldova
 Montenegro Mozambique Namibia Nicaragua Northern Ireland Panama Qatar Rwanda Syria Tanzania Trinidad and Tobago Wales
 Zimbabwe Bolivia Iceland Jamaica Kenya Lithuania Macao Morocco Paraguay Puerto Rico Uganda Uruguay Zambia Algeria
 Cyprus Estonia Ethiopia Hungary Myanmar Palestinian Territories Peru Poland Slovakia Slovenia Somalia Sudan
 Tunisia Yemen Guatemala Libya Oman Switzerland Venezuela Bosnia and Herzegovina Bulgaria Croatia
 Iraq Mongolia Nepal Nigeria Singapore Chile Colombia Denmark Ireland Lebanon Romania
 Serbia Ukraine Austria Belgium Czech Republic Greece Hong Kong New Zealand
 Philippines Russia Vietnam Jordan Norway Scotland Sri Lanka Malaysia
 Netherlands Argentina Finland Israel South Africa United Arab Emirates
 Bangladesh Japan Thailand Egypt Sweden Taiwan Indonesia Mexico
 Portugal Italy Saudi Arabia South Korea France Spain China Iran
 Turkey Australia Pakistan Canada Germany England Brazil
 United Kingdom India United States





National Portrait Gallery
London, UK
Oil on canvas, 1863

Ada Lovelace (1815-52) is widely regarded as **the first computer programmer and the first person to grasp the potential for computers outside mathematics**. She was the daughter of the poet Lord Byron and his wife Annabella Milbanke. Raised by her mother, later a renowned educationalist, Lovelace became famous in mathematics at a time when few women received any extended education. Charles Babbage became her friend and mentor, affectionately calling her the 'Enchantress of Numbers'. Together they became pioneers of computing with the idea of a mechanical calculator they called the Analytical Engine. **Various awards have been set up in her honour and the achievements of women in Science, Technology, Engineering and Mathematics (STEM) roles are now celebrated annually on Ada Lovelace Day.**

为什么要学这门课？

- 了解ML和PDA基本概念
- 学习几种基础的ML方法
- 读一本英文教科书，掌握ML的英文专业术语
- 探索利用ML方法解决实际问题...
- 为进一步从事AI领域研究奠定基础
- 作为中国TOP1大学的学生，与美国TOP1大学学生学习相同的课程😊

Inspiring!

Challenging!

Fulfilling!

内容及进度安排（48学时）

章节顺序	章节名称 Chapters	课时 Hours
1	引言：机器学习及数据分析预测的基本概念 Machine Learning for Predictive Data Analytics	3
2	从数据到洞察到决策 Data to Insights to Decision	2
3	数据探索 Data Exploration	3
4	基于信息的学习 Information-based Learning	5
5	基于相似性的学习 Similarity-based Learning	4
6	基于概率的学习 Probability-based Learning	6
7	基于误差的学习 Error-based Learning	5
8	深度学习 Deep Learning	6
9	评估 Evaluation	5
10	案例分析 Case Study	3
11	总结：应用于数据分析预测的机器学习技术 the Art of Machine Learning for Predictive Data Analytics	3

考核方式

- 平时成绩 50%
 - 习题 20%
 - 论文+报告 30%
- 期末考核成绩 50%

Requirements/Grading:

Term Assessments:

- Exam(s) given during term - 25%
- Problem sets - 15%
- Programming assignments - 10%

Final Assessments:

- Final scheduled exam - 50%

	2019	2020	2021	2022	2023	2024	2025
平均分	91.9	90.5	88.8	88.1	90.3	86.6	90.7
优秀率	100%	88%	76%	54% *25% 选 择P/F制	73%	58.8%	73%

2019 年 PROJECT 选 题

- 精品旅行服务成单预测
- 基于强化学习的FPGA深度学习加速器逻辑综合优化
- **Classification and clustering algorithm applied in SDSS catalogue**
- 性别对择偶偏好的影响
- **Spotify歌曲流行程度预测**
- 基于支持向量机的Autism Spectrum Disorder(ASD)诊断
- 基于价格买卖和追市行为的股票择时模型
- **Forecasting of Stock Price Index Based on Several Statistical Learning Method**
- 基于住房特征的技能水平预测
- 预测欧洲某连锁药店在7-9月的销售情况
-

2020 年 PROJECT 选 题

- 基于大数据和传统数据结合的北京房价预测模型
- 基于机器学习的大类资产配置策略研究
- Prediction of Combined Postoperative Outcomes after Surgery in Children Diagnosed with Congenital Coarctation of the Aorta, A Multivariable Logistic Regression Model with Lasso
- 利用宏观经济指标预测A股各行业指数月度收益率
- 心血管疾病预测的机器学习报告
- 机器学习方法在中文文本分级阅读任务中的应用尝试
- NBA球员进攻特性的聚类与分析
- 模型解释与数据、业务理解——以某电子竞技项目为例
- 基于新闻分类的不同 Machine Learning 方法比较
- 新冠疫情因素对股票策略的影响研究
- A Country-Level Prediction Model of Coronavirus Infective Rate
- The Prediction of Songs' Popularity Based on Machine Learning
- 基于机器学习检测社交媒体论坛中的负面情绪和自杀倾向
- 基于用户行为的个性化游戏推荐系统
- Stock Price Prediction using Machine Learning Algorithms
- 基于机器学习模型的债券违约因素探究
-

2021 年 PROJECT 选题

- 根据音乐特征进行音乐类型分析
- 根据NBA球员新秀赛季数据预测该球员的职业寿命超过五年的概率
- 房屋租金预测研究
- 利用数据挖掘预测学生成绩
- 利用数据预测二手车售价
- 基于Markov Chain的随机旋律生成研究
- 森林火灾的影响因素分析
- 预测部分城市房价
- 基于Transformer+LSTM的股票价格预测网络
- 基于机器学习的A股高频因子提取与整合
- 关于太阳辐射的预测
- NBA球星当日得分预测
- 基于Freesound数据集的音频标签（与音高预测）

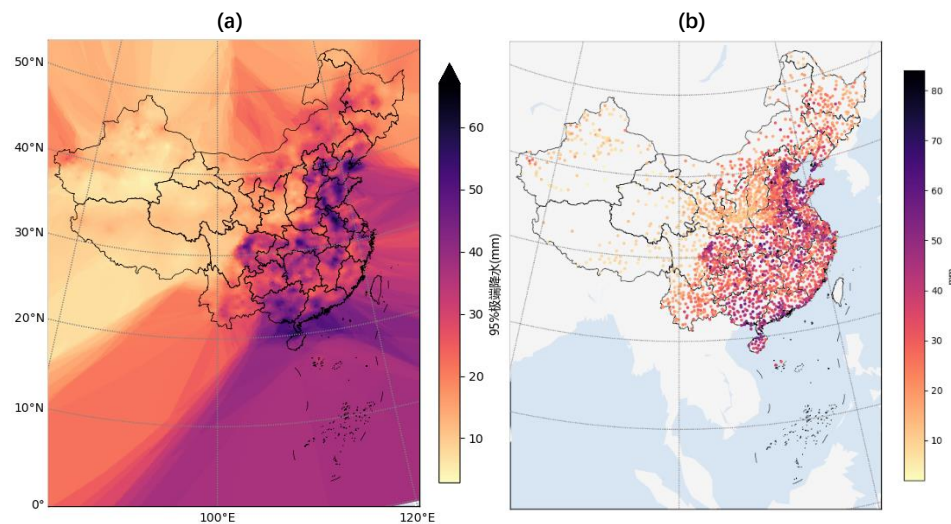


图6 (a) K近邻算法95%极端降水阈值插值分布图, 时间分辨单位10min, k=7
(b) 单站95%极端降水阈值分布, 时间分辨单位10min

- 英雄联盟排位赛各因素对胜率的影响
- 不同机器学习模型在股票收益预测的性能比较
- 基于分钟降水资料的中国地面极端降水研究
- Time-series forecasting in real cryptocurrency market: explore the extent of market prediction possibilities
- Heart Failure Prediction
- 依据多级神经网络和时序分割的手术影像识别
- Detecting Diabetic Retinopathy with U-Net
- 高铁列车4G信号强度预测
- 基于客户行为分析酒精饮料潜在客户群

机器学习算法在声学成像阵列中的应用研究

电影评论的文本情绪预测分析

基于封面信息的书籍分类

基于深度学习的股价预测研究

基于Bert模型的股票期权价格涨跌预测

基于机器学习的中国人肥胖因素的研究

基于深度学习的遥感影像分类

基于NBA历史数据预测球员职业生涯

基于深度学习的图像识别

基于RealSR的图像超分辨率网络实践

关于消费者网购是否会将商品加入购物车的预测

基于深度学习的中文谜语问答

基于医学图像的疾病预测系统

使用LSTM模型对股票的收益预测

基于CFPS数据的中国家庭经济状况预测

基于深度学习的音乐主旋律提取与旋律生成

美国纽约共享单车使用影响因素分析

基于汉语词典释义文本的词性预测

基于机器学习的日经股票收益率预测

基于博弈论和迁移学习的电动汽车电池数据价值计算方法

基于各因素预测2022世界杯比赛结果

	K近邻算法	贝叶斯网络	逻辑回归	支持向量机	Sequential 模型
第一名	法国	阿根廷	法国	法国	法国
第二名	阿根廷	法国	阿根廷	阿根廷	阿根廷
第三名	克罗地亚	克罗地亚	克罗地亚	克罗地亚	克罗地亚
第四名	摩洛哥	摩洛哥	摩洛哥	摩洛哥	摩洛哥

基于机器学习模型的商业电影票房预测

基于分钟级人体活动数据的抑郁状态分类和MADRS评分预测

探究不同机器学习模型在基于评分的推荐算法中的工作效果

机器学习在金融反欺诈中的应用

面部关键点检测

利用机器学习模型进行天气预测（包括气温、风向风速以及降水）

利用数据挖掘预测STEAM平台上游戏的评价

基于机器学习算法预测电影票房及综合评分

基于BP神经网络开发的简易自动消息回复系统

关于行人轨迹的预测问题

基于机器学习和指纹匹配法的室内WIFI信号定位研究

动漫作品偏好分析与推荐

基于CNN（卷积神经网络）的波束选择

基于机器学习结合时事热点的股价预测

探索影响预期寿命的因素

社交媒体有害评论分类任务

基于多模型的人类活动行为分类

预测英国足球超级联赛比赛结果

一个简易的英语作文多维度评估模型

机器学习视角下的《诗经》分韵问题

买入大单对价格的冲击

基于深度学习模型的本薯叶疾病类型预测

01

背景介绍

定位

- 室外定位：卫星导航系统
- 室内定位：？

WIFI 信号指纹定位

空间位置 ↔ 无线信号特征：指纹

训练过程：现场勘测，位置采样

服务过程：指纹上传，特征匹配

冗余？噪声？信息缺失？



图源：Song, X., et al. (2019). "A Novel Convolutional Neural Network Based Indoor Localization Framework With WiFi Fingerprinting." IEEE Access 7: 110698-110709.

2023年PROJECT选题

- 疾病预测与诊断的机器学习研究
- 基于GAN的图像风格转换器
- 基于机器学习对晋江网络文学进行热度预测
- 基于就餐指数分析的北京大学食堂就座率预测
- 脑卒中预测
- 基于贝叶斯网络的南海人工岛礁降水预测分析
- 北京市房价的分析预测
- Singlish or English? 基于机器学习和大语言模型的“新式英语”语言鉴别研究
- 基于用户特征的音乐偏好预测与推荐
- 由引文网络出发的科学出版物类别预测
- 根据CMI睡眠手环数据预测睡眠状态 (kaggle竞赛)
- Transformer时间序列选股策略
- 世界一级方程式锦标赛(F1)比赛结果预测
- 2023-2024年度北京市(及邻近地区)初雪时间预测
- 基于机器学习的人脸识别与分类
- 根据图片进行字体类型提取预测与生成
- 对酒店预订价格的预测
- 探究城市交通拥堵的影响因素
- A股上市公司行动事件对股票价格影响
- 人类风险决策建模中不同机器学习方法的比较评估
- 基于卷积神经网络的引力波信号探测图样识别
- 机器学习对于地震波噪音和事件的分类处理

- 基于机器学习的投资组合研究
- 基于机器学习预测2024-2025年欧洲冠军联赛结果
- 肥胖风险的多类别预测 Multi-Class Prediction of Obesity Risk
- 利用多模型对AirBnb房屋价格的预测
- Assessing the Susceptibility of Thermokarst Lakes Induced by Permafrost Degradation in the Qinghai-Tibet Plateau Using Machine Learning Methods
- 电影票房预测
- 基于机器学习对公共论坛中文本情绪倾向的预测
- 强化学习在CNN加速器RRAM阵列排布中的应用
- 根据Leetcode网站上题目描述及特征预测问题是否会在faang公司面试中出现
- 基于社交媒体发言数据的舆论情感趋势预测
- 基于机器学习的卡路里消耗预测
- 利用历史空气质量数据以及相关影响因素预测未来特定区域的空气质量水平
- 基于深度学习的动漫作品热度预测
- 基于点云深度学习网络的牙齿磨损评估
- 宠物领养情况预测
- 基于人类对局统计对围棋中“变招”的评价
- 用机器学习进行房价预测
-

- 基于机器学习预测2025年TGA年度游戏
- 探究惩罚因子与核函数对支持向量机分类效果的影响
- 基于机器学习解码人类心理意象的尝试
- 基于用户与广告特征的购买意向预测
- Spotify歌曲情绪分类
- 基于机器学习的音乐billboard排名预测
- 基于机器学习对国际武装冲突风险预测
- “七情百态”：静态图像与动态视频面部表情识别的对比研究 ——
基于机器学习与深度学习
- 基于机器学习方法的股票市场基金经理画像
- 基于宏观数据的大类资产配置模型
- 持续“买入”？——基于FinBERT与机器学习的情感悖论与风险预测
- 基于机器学习的银行存款预测
- 电商用户购买行为预测
- Using Machine Learning to Analyse Factors of Viral Beauty Products
-

作品1：记忆的复原——基于深度学习的老照片修复系统

作品2：基于稠密连接网络与生成式对抗网络的医学影像超分处理系统



2021年(第14届)中国大学生计算机设计大赛

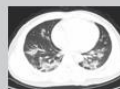
The 14th Chinese Collegiate Computing Competition (4C2021)

人工智能应用（百度杯）/ 数媒动漫与短片专业组

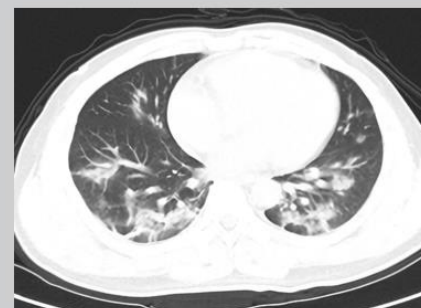
2021年8月14日-17日、8月26日-30日

主办：中国大学生计算机设计大赛组委会
协办：中国大学生计算机设计大赛南京决赛赛区组委会
承办：三江学院（南京）
协办：百度在线网络技术有限公司

低分辨率图像



4倍超分的高分辨率图像



- https://ai100.stanford.edu/sites/g/files/sbiybj18871/files/media/file/AI100Report_MT_10.pdf

2021年9月发布

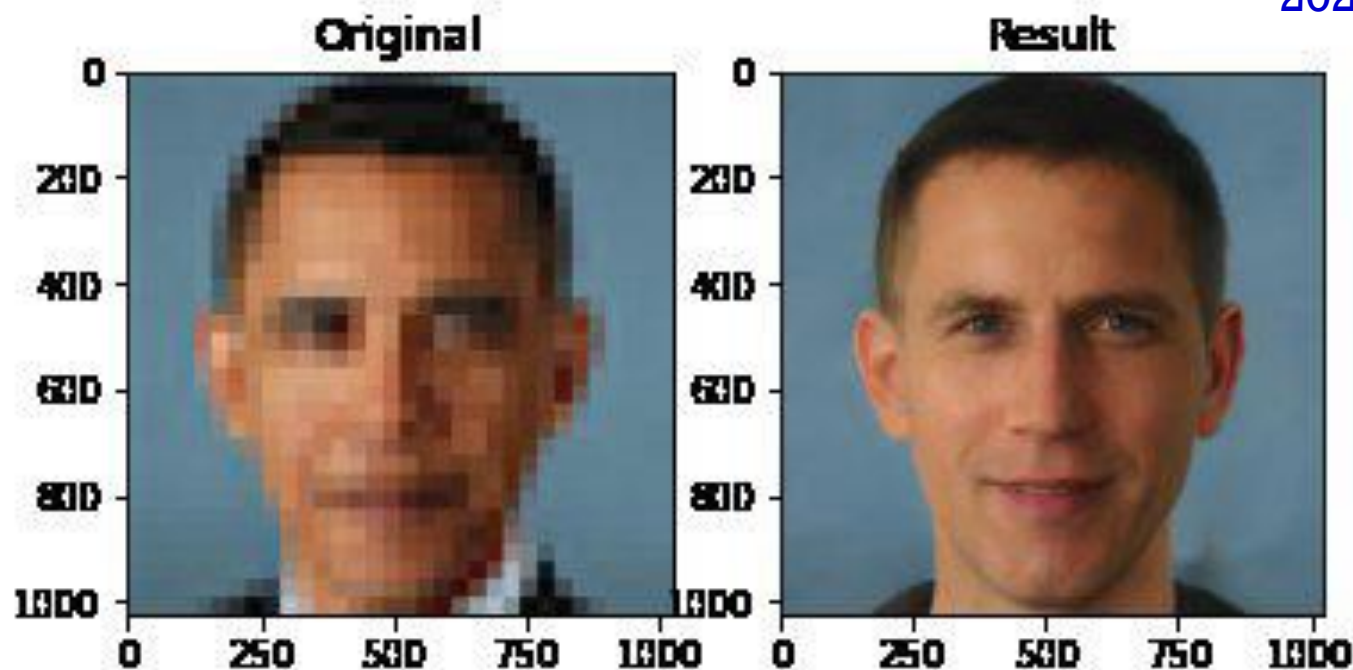


Image-generation GANs can be used to perform other tasks like translating low-resolution images of faces into high resolution images of faces. Of course, such a transformation **is not recovering missing information so much as it is confabulating details that are consistent with its input**. As an example, the PULSE system tends to generate images with features that appear ethnically white, as seen in this input image of former US President Barack Obama.

From: <https://www.theverge.com/21298762/face-depixelizer-ai-machine-learning-tool-pulse-stylegan-obama-bias>





遗彩重现 AI辅助考古图像修复系统

2025年（第18届）中国大学生计算机设计大赛



方法创新——首创“分源-拼接-修复”三步修复法

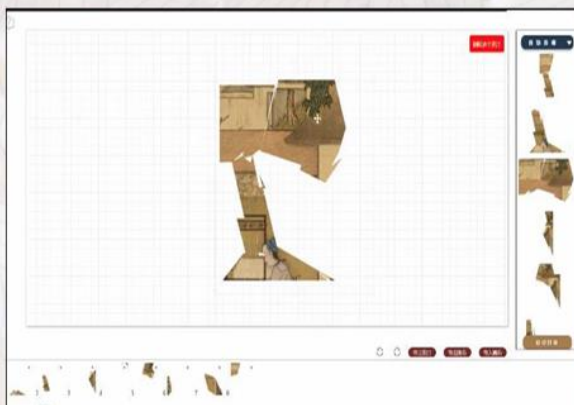
分源

多源碎片分源解决方案



拼接

内容主导的拼接解决方案



修复

内部、外部修复解决方案



依托文物数据集，采用模式识别技术，对碎片进行**AI来源分辨**，降低拼接难度。

结合AI技术，给出碎片的**虚拟匹配**方案，提高拼接效率。

建造一体化数字文物修复平台，**修补破损**图像，确保方案的准确性。



2025年（第18届）中国大学生计算机设计大赛

The 18th Chinese Collegiate Computing Competition (4C2025)

南京东决赛区





2026年（第19届）中国大学生计算机设计大赛

The 19th Chinese Collegiate Computing Competition (4C2026)

南京东1决赛区

人工智能应用（新大陆杯） | 数媒游戏与交互设计

江苏·无锡 | 2026年8月16日-20日



梯净智驱

基于多模态感知的自适应三段式机器人

全国首款
基于 AI 全域感知的三段式
多地形智能楼梯清扫机器人

行业首创
“三段自适应” 闭环式
多地形楼梯清扫作业系统



- 关于本课程
- AI历史回顾
- Machine Learning for Predictive Data Analytics

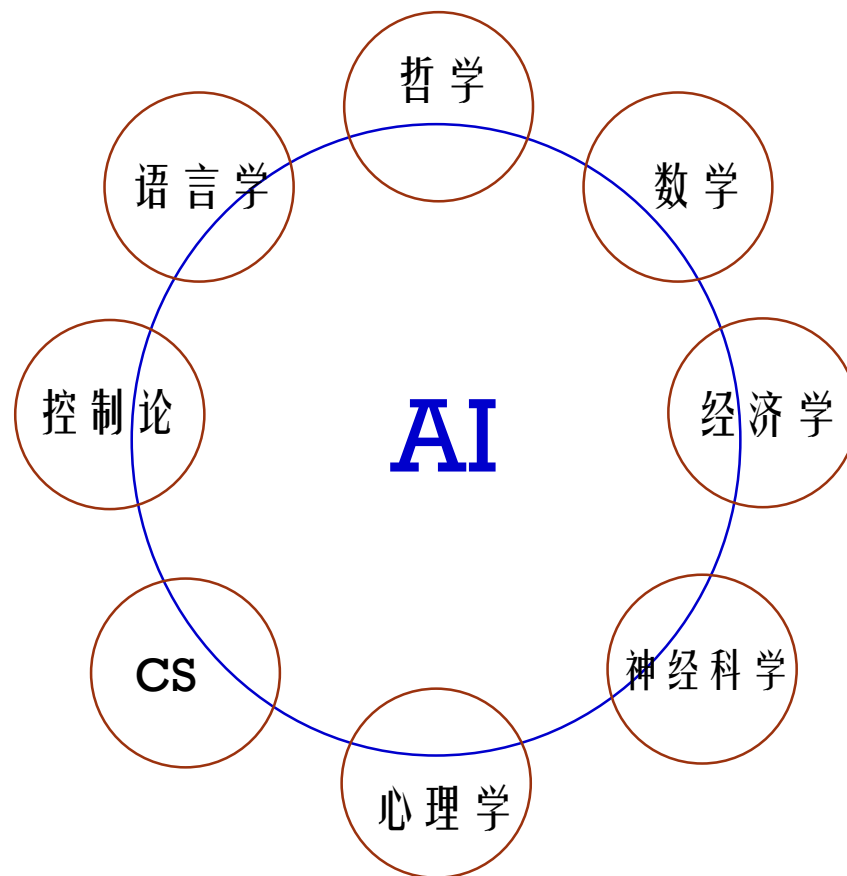
■ 人工智能（ AI-Artificial Intelligence ）涉及非常广泛的领域：

- （ 1 ）自然语言处理 (natural language processing)
- （ 2 ）知识表示 (knowledge representation)
- （ 3 ）自动推理 (automated reasoning)
- （ 4 ）机器学习 (machine learning)
 - 深度学习 (deep learning)
- （ 5 ）计算机视觉 (computer vision)
- （ 6 ）机器人学 (robotics)

Interdisciplinary

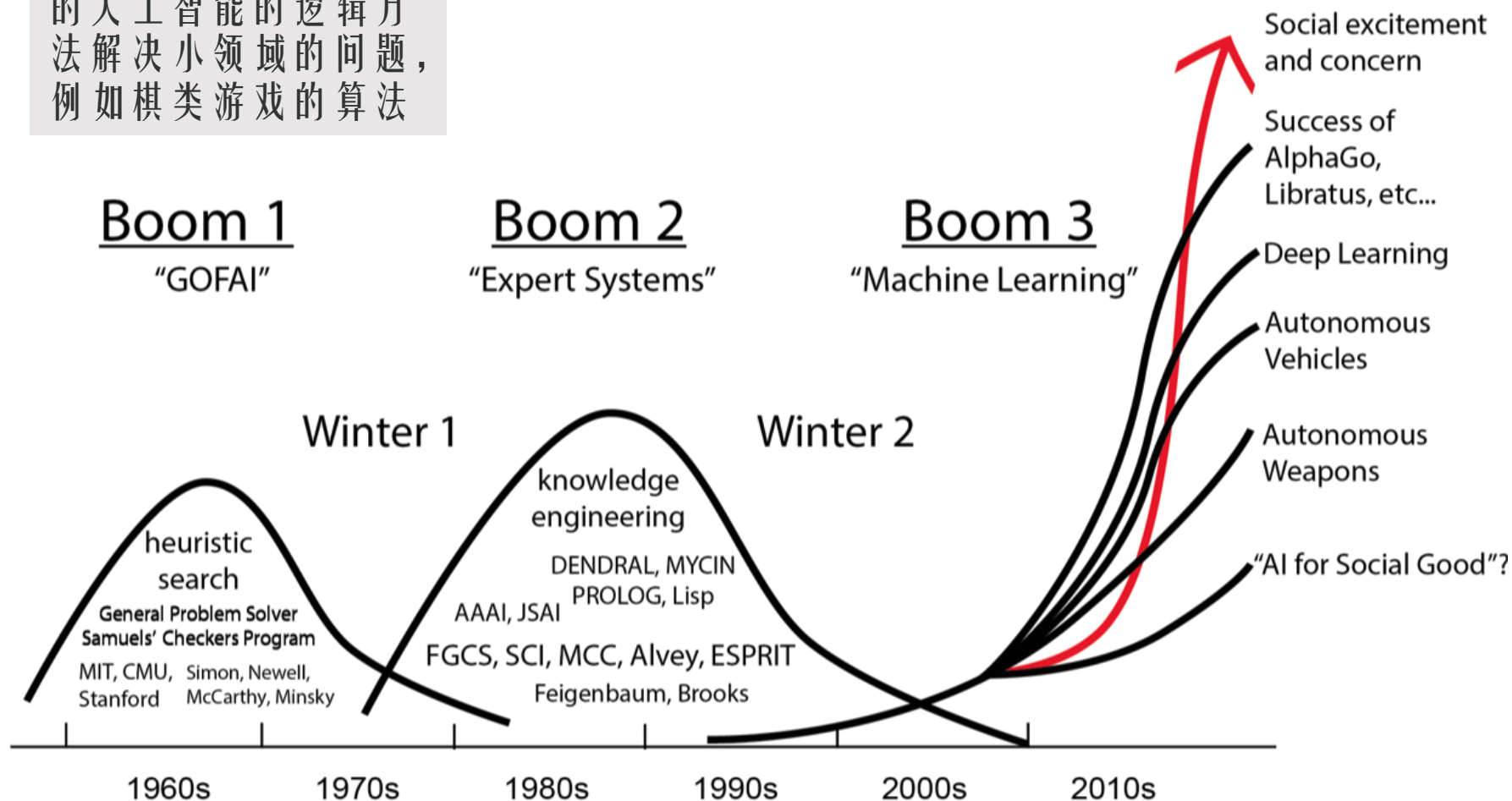
■ 人工智能的基础：

- 哲学
- 数学
- 经济学
- 神经科学
- 心理学
- 计算机科学
- 控制论
- 语言学
-



历史回顾

GOFAI 泛指用最原始的人工智能的逻辑方法解决小领域的问题，例如棋类游戏的算法



- AI's official birth: Dartmouth, 1956

“An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. ***We think that a significant advance can be made if we work on it together for a summer.***”

John McCarthy and Claude Shannon

Dartmouth Workshop Proposal

1956 Dartmouth Conference: The Founding Fathers of AI



John MacCarthy



Marvin Minsky



Claude Shannon



Ray Solomonoff



Alan Newell



Herbert Simon



Arthur Samuel



Oliver Selfridge



Nathaniel Rochester

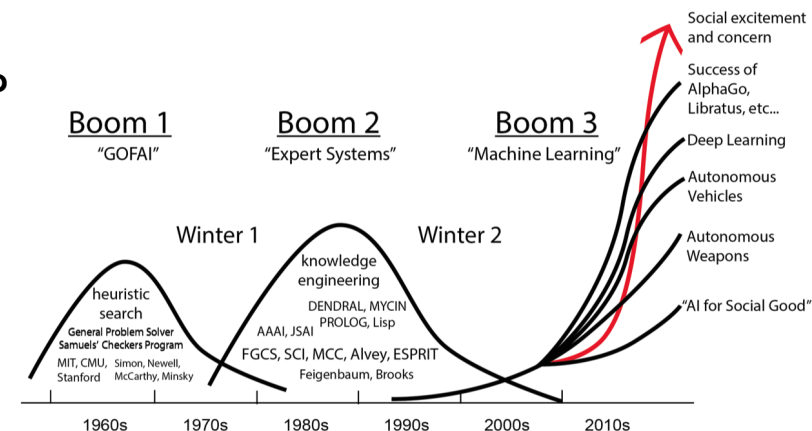


Fred Moore

@稀土掘金技术社区
Tree

历史回顾

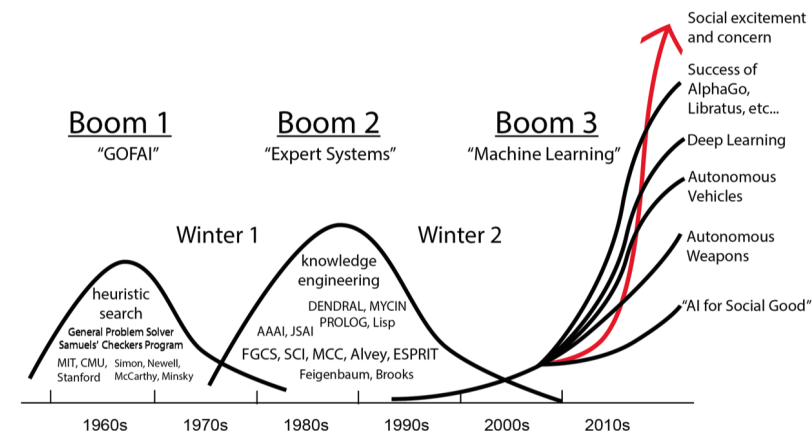
- **1940-1950: Early days**
 - 1943: McCulloch & Pitts: Boolean circuit model of brain
 - 1950: Turing's "Computing Machinery and Intelligence"
- **1950—70: Excitement!**
 - 1950s: Early AI programs: chess, checkers program, theorem proving
 - **1956: Dartmouth meeting: "Artificial Intelligence" adopted**
 - 1965: Robinson's complete algorithm for logical reasoning
- **1970—90: Knowledge-based approaches**
 - 1969—79: Early development of knowledge-based systems
 - 1980—88: Expert systems industry booms
 - 1988—93: Expert systems industry busts: "AI Winter"
- **1990—2012: Statistical approaches + subfield expertise**
 - Resurgence of probability, focus on uncertainty
 - General increase in technical depth
 - Agents and learning systems... "AI Spring"?
- **2012—__: Excitement?**
 - Big data, big compute, neural networks
 - Some re-unification of sub-fields
 - AI used in many industries



历史回顾

■ AI everywhere...

- Search engines
- Route planning, e.g. maps, traffic
- Logistics, e.g. packages, inventory, airlines
- Medical diagnosis, machine diagnosis
- Automated help desks
- Spam / fraud detection
- Smarter devices, e.g. cameras
- Product recommendations
- Assistants, smart homes
- ... Lots more!



AI is the new electricity



- 100 years ago, electricity transformed every industry.
- AI is bringing an equally big transformation.

微信@吴恩达之城

Opinion: We've reached a turning point with AI, expert says

<https://edition.cnn.com/2023/05/31/opinions/artificial-intelligence-stuart-russell/index.html>

By Jessica Chia and Bethany Cianciolo, CNN

Updated 12:46 PM EDT, Tue September 5, 2023



Editor's Note: This is the first installment of a new CNN Opinion project dedicated to examining the potential and the risks of artificial intelligence. **"Our AI future: promise and peril"** will explore how AI will affect our lives, the way we work and how we understand ourselves.

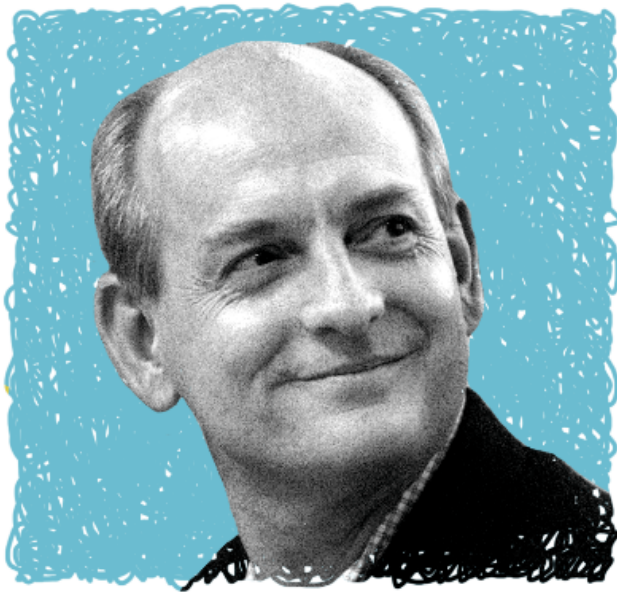
(CNN) — Ever since ChatGPT was released last November, artificial intelligence has been thrust into the spotlight, sparking enormous excitement and debate over the possibilities.

ChatGPT, along with other AI tools like DALL-E, Midjourney, Stable Diffusion and Bard, racked up millions of users, who utilized them to write emails, plan their vacations, impersonate musicians, produce campaign ads and even design buildings.

While tech giants like Bill Gates have touted the possibility that artificial intelligence can reduce global inequality or fight climate change, the technology has also prompted a lot of fear and anxiety: Will AI replace millions of jobs? Will disinformation become even more widespread? Will general purpose AI — AI that is as capable as humans — eventually take over the world?

We talked to Stuart Russell, a computer science professor at the University of California, Berkeley who co-authored the textbook, "Artificial Intelligence: A Modern Approach," about the promises and risks of AI, and whether it's possible to ensure it remains safe and within our control.

Russell has called for rebuilding AI on a different foundation to ensure our control over the technology — but that doesn't solve the potential issue of AI systems falling into the hands of malign forces.



CNN/Peg Skorpinski

“...even though we may understand how to build perfectly safe general purpose AI, what’s to stop Dr. Evil building general purpose AI that’s going to destroy the world?”

Stuart Russell

Russell also worries about humans becoming too dependent on the technology, and then losing “the incentive to learn and to be capable of anything. And that, I think, would be another form of catastrophe,” he said.

Read our conversation below, which has been edited and condensed for clarity.

CNN: As someone who has been in the field and studied AI for many decades, how does it feel to have AI be such a huge topic of interest all of a sudden?

Opinion: Here are the jobs AI will impact most

<https://edition.cnn.com/2023/09/05/opinions/artificial-intelligence-jobs-labor-market/index.html>

Opinion by [Bethany Cianciolo](#) and [Jessica Chia](#), CNN

Updated 10:09 AM EDT, Tue September 5, 2023



Editor's Note: This is the second installment of a CNN Opinion project dedicated to examining the potential and the risks of artificial intelligence. **"Our AI future: promise and peril"** explores how AI will affect our lives, the way we work and how we understand ourselves.

(CNN) — "AI is coming for your job."

That's just one variation of the many headlines you've probably seen ever since ChatGPT exploded in popularity and won the world's attention late last year. But is it true?

Back in April, [Dropbox](#) announced it was cutting 500 employees. In May, outplacement firm Challenger, Gray & Christmas let go of almost 4,000 people. And in July, the [founder](#) of an e-commerce startup said he laid off 90% of his support team. The common reason cited? You guessed it: artificial intelligence.

[Goldman Sachs economists](#) have estimated that 300 million full-time jobs across the globe could be automated in some way by the newest wave of AI, with up to a quarter of all jobs being completely done by AI.

Indeed, large language models like [ChatGPT](#) have demonstrated a pretty remarkable ability to write code, offer detailed instructions for different tasks, pass a law school bar exam, and even express [empathy](#) when answering medical questions. And while this technology has the potential to cause widespread

▶ Opinion: Here are the jobs AI will impact most

▶ Medicine

Erich S. Huang: Do you want AI to be your doctor?

Regina Barzilay: AI can shift late-stage treatment to prevention

▶ Law

Daniel W. Linna Jr.: AI is changing the skills lawyers need

▶ Art

Refik Anadol: AI will force artists to push their imaginations even further

▶ Retail

Adam Elmachtoub: AI will help grocery stores set inventory for game days

Nisreen Ameen: Not sure what sunglasses suit you? AI can analyze your face and decide

▶ Film

Theodore Kim: Get ready for the rise of the mockbuster

▶ Tech

Eirini Kalliamvakou: A big win for software developers — and society

▶ Education

Ashok Goel: AI will fully power universities in the next few years

Kristen DiCerbo: AI will allow teachers to be more human

▶ Agriculture

Alireza Pourreza: When it comes to AI, farmers will need to strike a balance



CNN/Adobe Stock

“While AI might simulate compassion, beyond chatbot interactions, AI cannot truly empathize or be emotionally anticipatory the way human clinical professionals can.”

Erich S. Huang

Medicine

Erich S. Huang: Do you want AI to be your doctor?

Erich S. Huang is the head of Clinical Informatics at Verily and former chief data officer for quality at Duke Health and assistant dean for Biomedical Informatics at [Duke University School of Medicine](#).

I’ve worked on AI in biomedicine for my entire career. I absolutely believe in its potential and know that AI will certainly be a part of medicine as we work to improve and further personalize care. AI can win back time and space for clinicians and help to reduce the administrative burden that is mostly tangential to direct patient care.

How often do you notice your doctor or nurse looking at a screen rather than looking at you? Several hours and thousands of mouse clicks a day are laboriously devoted to entering data into an electronic health record. Many health care person hours are spent on billing and reimbursement rather than patient care. These represent low-hanging opportunities for automation that will help your doctor be more present for you.

But AI should not take away clinical jobs. While AI might simulate compassion, beyond chatbot interactions, AI cannot truly empathize or be emotionally anticipatory the way human clinical professionals can.

Law

Daniel W. Linna Jr.: AI is changing the skills lawyers need

Daniel W. Linna Jr. has a joint appointment at [Northwestern's](#) Pritzker School of Law and McCormick School of Engineering. Dan's research and teaching focus is on using AI for legal services and the regulation of AI in society. Previously, Dan was a litigator and equity partner at Honigman, a large law firm, and an IT manager, developer and consultant.



CNN/Adobe Stock

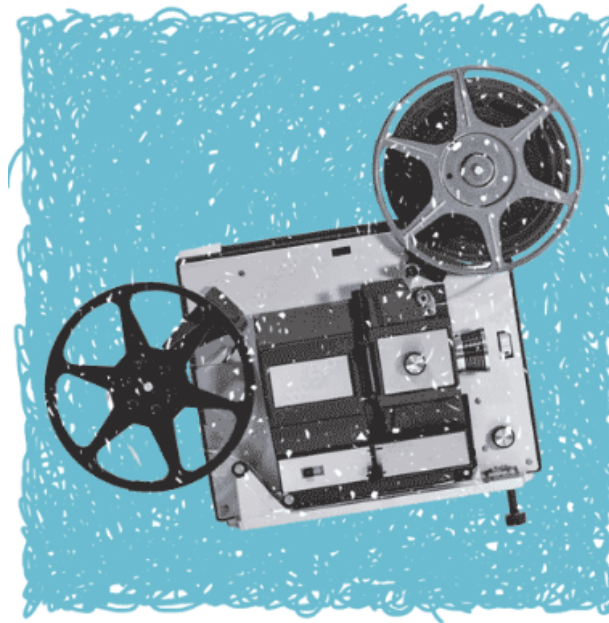
“...in a business dispute for nonpayment of goods, an AI system could predict the likelihood of success and create initial drafts of legal briefs ... based on the AI system’s analysis of the judge’s past written decisions.”

Daniel W. Linna Jr.

For lawyers, this means that AI can automate or augment many legal tasks that they perform. Most lawyers spend a lot of time finding applicable laws, organizing information, spotting common issues, performing basic analysis and drafting formulaic language in emails, memos, forms, contracts and briefs. AI systems will be able to do this faster, cheaper and better. Large language models, like those behind ChatGPT, have significantly increased the capabilities of these systems. Established legal information providers and many [startups](#) are rapidly developing and releasing [AI systems](#) that are “fine-tuned” or specialized for [legal tasks](#).

Unsurprisingly, AI is changing the skills lawyers need. To responsibly use AI, they will need a functional understanding of the technology to evaluate the benefits and risks of using it, such as how it might fail and the ways in which it might be biased or unfair.

Lawyers will also need to exercise judgment to tailor an AI system’s output for specific situations. For example, in a business dispute for nonpayment of goods, an



CNN/Adobe Stock

“...the race to the bottom sparked by AI is going to see this genre explode, where every big budget movie gets automatically ingested and imitated.”

Theodore Kim

Film

Theodore Kim: Get ready for the rise of the mockbuster

Theodore Kim is a professor of computer science at [Yale](#), a former senior research scientist for Pixar and a two-time Scientific and Technical Academy Award winner.

Moreover, if all this comes to pass, we should prepare for the golden age of the mockbuster. With no protections in place for writing or acting, the race to the bottom sparked by AI is going to see this genre explode, where every big budget movie gets automatically ingested and imitated.

The ChatGPT prompts will begin with, “Make me a ‘Mission Impossible,’ but different enough that I can’t get sued.” These AI-generated films will be a computational distillation of that episode of “30 Rock” where a Janis Joplin biopic that failed to secure her life rights was transformed into the more legally defensible “Jackie Joromp-Jomp” story.

As much as advocates breathlessly proclaim that AI will unleash new forms of creative expression, it excels most at imitation and interpolation, and may as well have been custom-made for mockbusters. We’re already seeing AI-generated trailers rip off Baz Luhrmann’s “The Great Gatsby” in a litigiously ambiguous manner. That’s not Leonardo DiCaprio, the argument goes. The hair color is different and the face is creepily blurrier.

Why are Hollywood **actors** and writers **on** strike?



Hollywood's **actors** and writers will join forces **on** the picket line from 14 July, after studios failed to reach a deal with the Screen Actors Guild - American Federation of Television and Radio Artists (SAG-AFTRA).

It is the first time the two unions have been **on** strike simultaneously since 1960.

Among SAG-AFTRA's 160, 000-strong ranks are many of the world's biggest stars. Hollywood's A-listers, from Tom Cruise to Angelina Jolie to Johnny Depp, are card-carrying union members.

Why are they striking?

The **industry-wide** shutdown follows a row about pay and the impact of **artificial** intelligence with film studios and streaming services.

```
main( )  
{  
    printf("hello, I
```

CNN

“The trajectory of the software industry includes many attempts to increase automation, and while that changed the nature of coding, it didn’t end the need for developers.”

Eirini Kalliamvakou

Tech

Eirini Kalliamvakou: A big win for software developers — and society

Dr. Eirini Kalliamvakou is a staff researcher at [GitHub Next](#), where she guides the team’s strategic prototyping efforts. Eirini has led the productivity research on GitHub Copilot, and has spoken extensively about developer productivity and happiness

More than 2,000 developers have reported that GitHub Copilot helps them stay in the flow, focus on more satisfying work and conserve mental energy. That’s a clear win for overall developer happiness.

As for how developers might use the time saved by AI? They can focus more on problem solving, software architecture and innovation. Development work is endless, and software companies and projects often have long backlogs of features, ideas and innovations that they never get to build because their resources are poured into the day’s work.

This increase in productivity could make a huge material difference — research shows that AI developer tools could boost global gross domestic product by more than \$1.5 trillion by 2030. Given the benefits, learning how to use these tools will soon become table stakes for software developers, who will become more fluent in prompting and interacting with AI. This will ultimately help democratize access to a career in software development and establish AI pair programming tools as part of the standard education and training in this field.



2 EXECUTIVE SUMMARY

Despite \$30–40 billion in enterprise investment into GenAI, this report uncovers a surprising result in that 95% of organizations are getting zero return. The outcomes are so starkly

- **Despite \$30–40 billion in enterprise investment into GenAI, this report uncovers a surprising result in that 95% of organizations are getting zero return.** The outcomes are so starkly divided across both buyers (enterprises, mid-market, SMBs) and builders (startups, vendors, consultancies) that we call it the GenAI Divide. Just 5% of integrated AI pilots are extracting millions in value, while the vast majority remain stuck with no measurable P&L impact. This divide does not seem to be driven by model quality or regulation, but seems to be determined by approach.
- Tools like **ChatGPT** and **Copilot** are widely adopted. Over 80 percent of organizations have explored or piloted them, and nearly 40 percent report deployment. But these tools primarily enhance individual productivity, not P&L performance. Meanwhile, enterprisegrade systems, custom or vendor-sold, are being quietly rejected. Sixty percent of organizations evaluated such tools, but **only 20 percent reached pilot stage and just 5 percent reached production.** Most fail due to brittle workflows, lack of contextual learning, and misalignment with day-to-day operations.

external agency use, particularly in back-office operations. Others cite improved customer retention and sales conversion through automated outreach and intelligent follow-up systems. These early results suggest that learning-capable systems, when targeted at specific processes, can deliver real value, even without major organizational restructuring.

- 关于本课程
- AI历史回顾
- Machine Learning for Predictive Data Analytics

CONTENTS

- Machine Learning for Predictive Data Analytics
 - [What is Predictive Data Analytics?](#)
 - What is Machine Learning?
 - How Does Machine Learning Work?
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- Predictive Data Analytics encompasses the **business** and **data processes** and **computational models** that enable a business to make **data-driven decisions**.

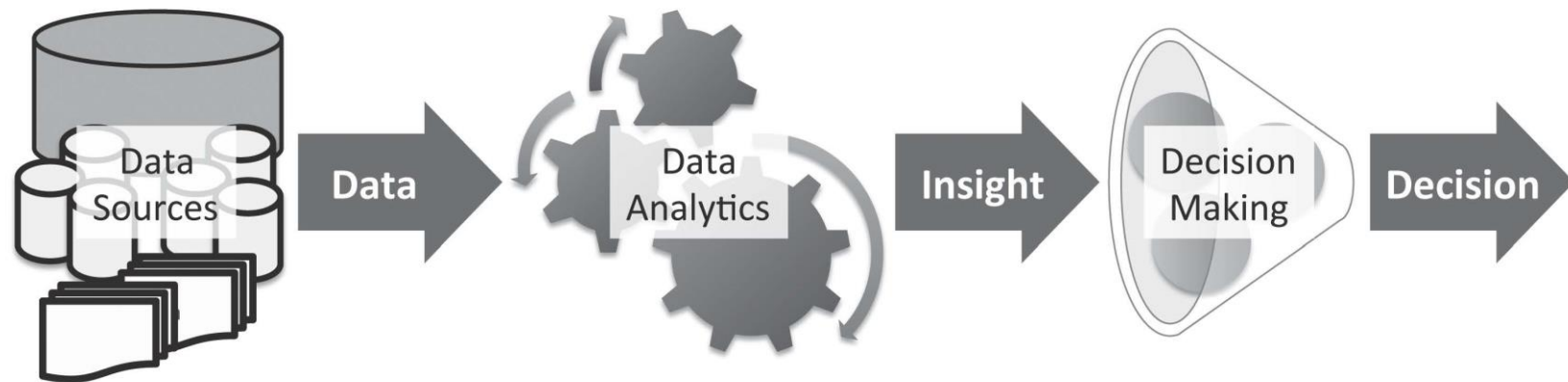


Figure 1.1: Predictive data analytics moving from **data** to **insights** to **decisions**.

EXAMPLE APPLICATIONS:

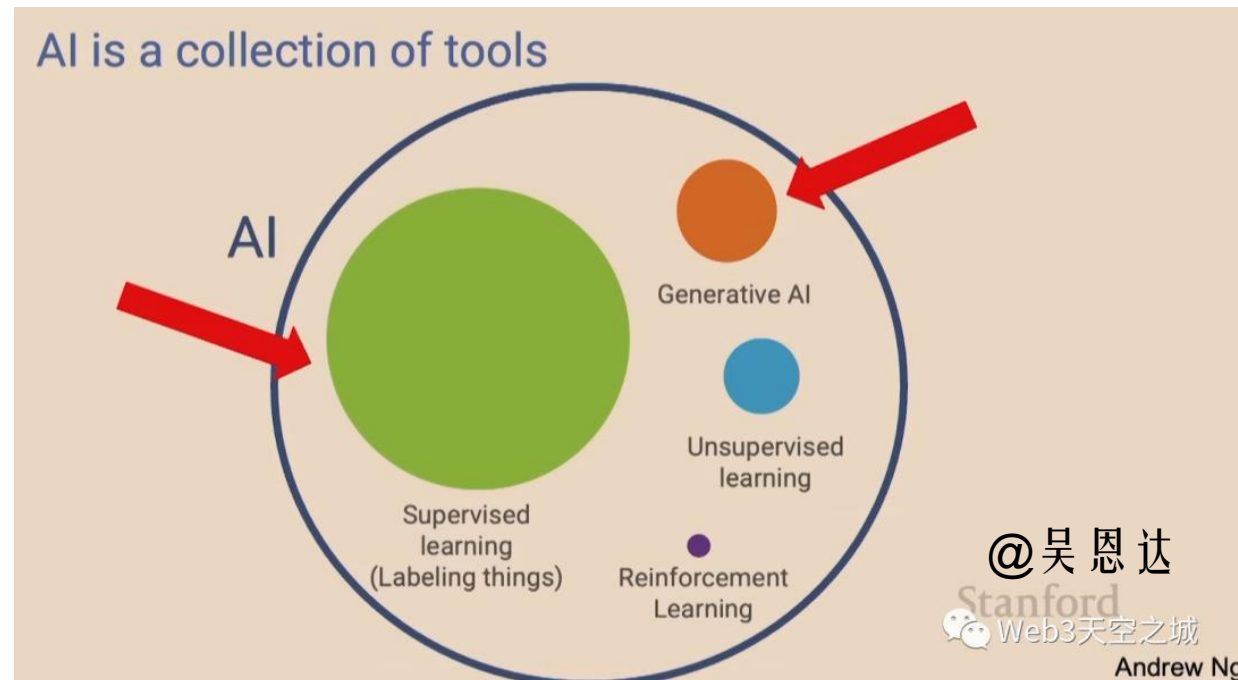
- Price Prediction
- Fraud Detection
- Dosage Prediction
- Risk Assessment
- **Propensity modelling**
- Diagnosis
- Document Classification
- ...

Successful applications of **propensity modeling** include predicting the likelihood of customers to leave one mobile phone operator for another, to respond to particular marketing efforts, or to buy different products.

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- (Supervised) Machine Learning techniques automatically learn a model of the relationship between a set of **descriptive features** and a **target feature** from a set of historical examples.



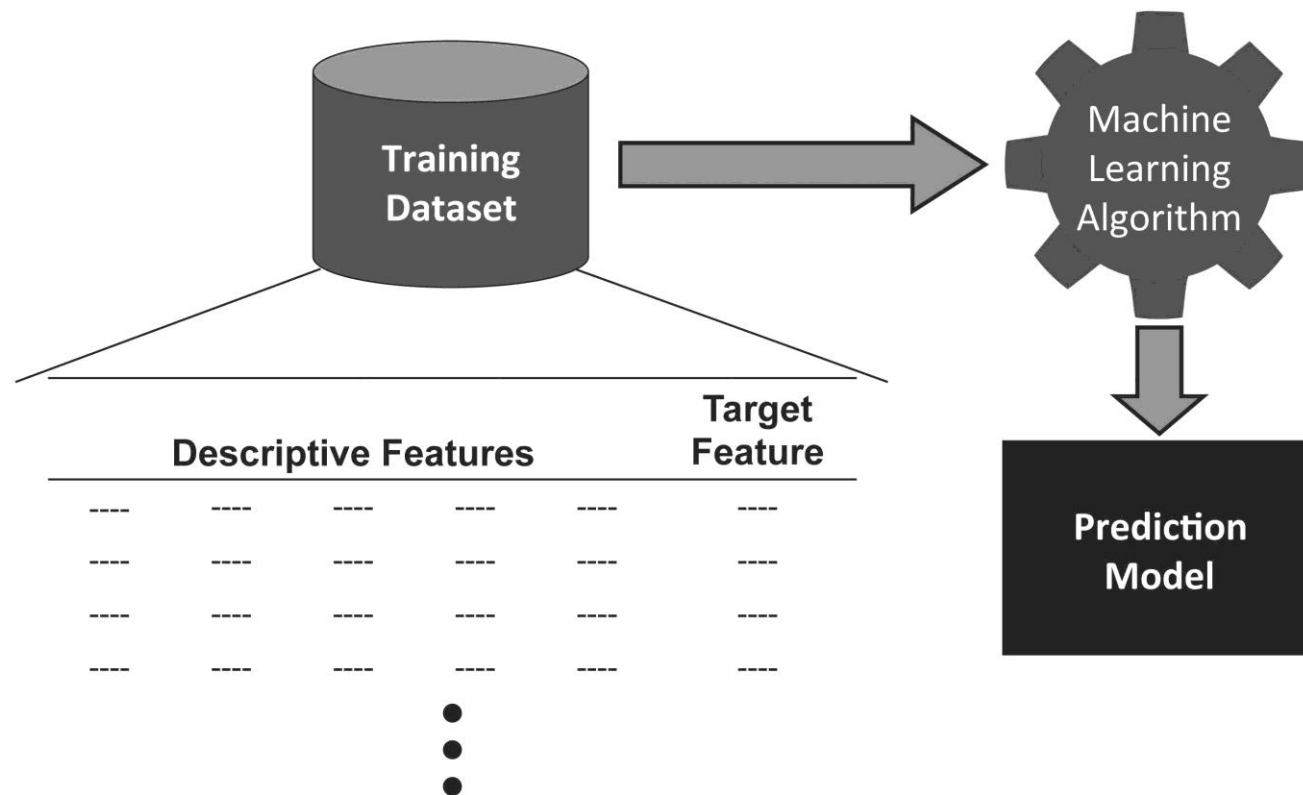


Figure 1.2(a): Using machine learning to induce a prediction model from a training dataset.

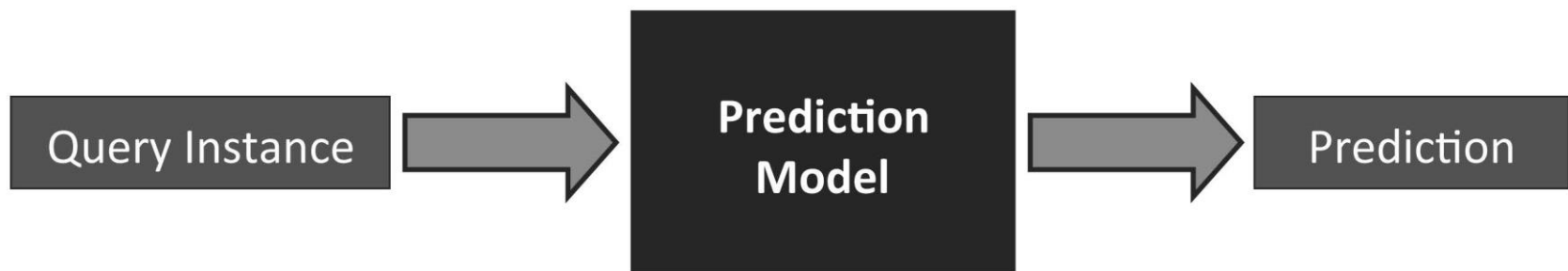


Figure 1.2(b): Using the model to make predictions for new query instances.

- A set of historical instances, or dataset, of **mortgages** that a bank has granted in the past.

Table 1.1: A credit scoring dataset

ID	OCCUPATION	AGE	LOAN-SALARY		OUTCOME
				RATIO	
1	industrial	34		2.96	repaid
2	professional	41		4.64	default
3	professional	36		3.22	default
4	professional	41		3.11	default
5	industrial	48		3.80	default
6	industrial	61		2.52	repaid
7	professional	37		1.50	repaid
8	professional	40		1.93	repaid
9	industrial	33		5.25	default
10	industrial	32		4.15	default

- What is the relationship between the **descriptive features** (OCCUPATION, AGE, LOAN-SALARY RATIO) and the **target feature** (OUTCOME)?

```
if LOAN-SALARY RATIO > 3 then
  OUTCOME='default'
else
  OUTCOME='repay'
end if
```

- This is an example of a **prediction model**
- This is also an example of a **consistent** prediction model
- Notice that this model does not use all the features and the feature that it uses is a **derived feature** (in this case a ratio): **feature design** and **feature selection** are two important topics that we will return to again and again.

- What is the relationship between the **descriptive features** and the **target feature** (OUTCOME) in the following dataset?

Table 1.2: A more complex credit scoring dataset

ID	Amount	Salary	Loan-Salary Ratio	Age	Occupation	House	Type	Outcome
1	245,100	66,400	3.69	44	industrial	farm	stb	repaid
2	90,600	75,300	1.2	41	industrial	farm	stb	repaid
3	195,600	52,100	3.75	37	industrial	farm	ftb	default
4	157,800	67,600	2.33	44	industrial	apartment	ftb	repaid
5	150,800	35,800	4.21	39	professional	apartment	stb	default
6	133,000	45,300	2.94	29	industrial	farm	ftb	default
7	193,100	73,200	2.64	38	professional	house	ftb	repaid
8	215,000	77,600	2.77	17	professional	farm	ftb	repaid
9	83,000	62,500	1.33	30	professional	house	ftb	repaid
10	186,100	49,200	3.78	30	industrial	house	ftb	default
11	161,500	53,300	3.03	28	professional	apartment	stb	repaid
12	157,400	63,900	2.46	30	professional	farm	stb	repaid
13	210,000	54,200	3.87	43	professional	apartment	ftb	repaid
14	209,700	53,000	3.96	39	industrial	farm	ftb	default
15	143,200	65,300	2.19	32	industrial	apartment	ftb	default
16	203,000	64,400	3.15	44	industrial	farm	ftb	repaid
17	247,800	63,800	3.88	46	industrial	house	stb	repaid
18	162,700	77,400	2.1	37	professional	house	ftb	repaid
19	213,300	61,100	3.49	21	industrial	apartment	ftb	default
20	284,100	32,300	8.8	51	industrial	farm	ftb	default
21	154,000	48,900	3.15	49	professional	house	stb	repaid
22	112,800	79,700	1.42	41	professional	house	ftb	repaid
23	252,000	59,700	4.22	27	professional	house	stb	default
24	175,200	39,900	4.39	37	professional	apartment	stb	default
25	149,700	58,600	2.55	35	industrial	farm	stb	default

ftb-first time buyer
stb-second time buyer

```
if LOAN-SALARY RATIO < 1.5 then
  OUTCOME='repay'
else if LOAN-SALARY RATIO > 4 then
  OUTCOME='default'
else if AGE < 40 and OCCUPATION = 'industrial' then
  OUTCOME='default'
else
  OUTCOME='repay'
end if
```

- The **real value** of machine learning becomes apparent in situations like this when we want to build prediction models from **large datasets** with **multiple features**.

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- Machine learning algorithms work by searching through a set of possible prediction models for the model that **best captures** the relationship between the descriptive features and the target feature.
- An obvious search criteria to drive this search is to look for models that are **consistent** with the data.
- However, because a training dataset is only a sample, ML is an **ill-posed** problem.

*An **ill-posed problem** is a problem for which a unique solution cannot be determined using only the information that is available.*

Table 1.3: A simple retail dataset

ID	BBY	ALC	ORG	GRP
1	no	no	no	couple
2	yes	no	yes	family
3	yes	yes	no	family
4	no	no	yes	couple
5	no	yes	yes	single

BBY - buys baby food

ALC - alcohol

ORG - Organic vegetable products

Table 1.4(a): A full set of potential prediction models before any training data becomes available.

BBY	ALC	ORG	GRP	M ₁	M ₂	M ₃	M ₄	M ₅	...	M _{6,561}
no	no	no	?	couple	couple	single	couple	couple		couple
no	no	yes	?	single	couple	single	couple	couple		single
no	yes	no	?	family	family	single	single	single		family
no	yes	yes	?	single	single	single	single	single		couple
yes	no	no	?	couple	couple	family	family	family	...	family
yes	no	yes	?	couple	family	family	family	family		couple
yes	yes	no	?	single	family	family	family	family		single
yes	yes	yes	?	single	single	family	family	couple		family

 $3^8 = 6,561$ possible prediction models!

the blanked out columns in the table indicate the models that are **not consistent** with the training data.

Table 1.4(b): A sample of the models that are consistent with the training data.

BBY	ALC	ORG	GRP	M ₁	M ₂	M ₃	M ₄	M ₅	...	M ₆ 561
no	no	no	couple	couple	couple	single	couple	couple		couple
no	no	yes	couple	single	couple	single	couple	couple		single
no	yes	no	?	family	family	single	single	single		family
no	yes	yes	single	single	single	single	single	single		couple
yes	no	no	?	couple	couple	family	family	family	...	family
yes	no	yes	family	couple	family	family	family	family		couple
yes	yes	no	family	single	family	family	family	family		single
yes	yes	yes	?	single	single	family	family	couple		family

$3^3 = 27$ possible prediction models!

- Notice that there is more than one candidate model left!
- It is because a single consistent model cannot be found based on a **sample** training dataset that ML is **ill-posed**.

- Consistency $\approx$ **memorizing** the dataset.
- Consistency with **noise** in the data isn't desirable.
- Goal: a model that **generalizes** beyond the dataset and that isn't influenced by the noise in the dataset.
- So what criteria should we use for choosing between models?

- **Inductive bias** the set of assumptions that define the model selection criteria of an ML algorithm.
- There are two types of bias that can be used:
 - restriction bias
 - A **restriction bias** constrains the set of models that the algorithm will consider during the learning process.
 - Multivariable linear regression with gradient descent
 - preference bias
 - A **preference bias** guides the learning algorithm to prefer certain models over others.
 - Iterative Dichotomizer 3 (ID3)
- Inductive bias is necessary for learning (beyond the dataset).

HOW ML WORKS (SUMMARY)

- ML algorithms work by searching through sets of potential models.
- There are two sources of information that guide this search:
 - the training data
 - the inductive bias of the algorithm

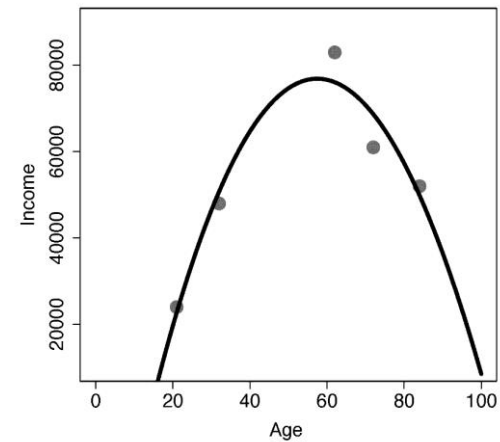
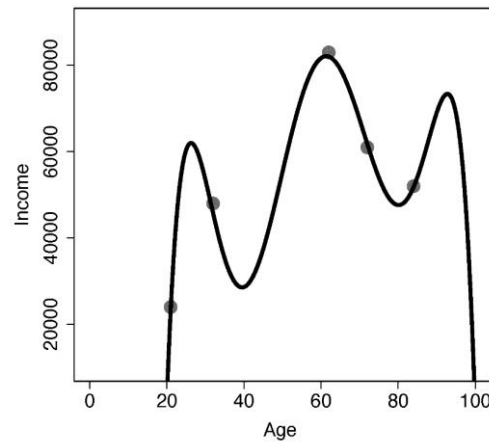
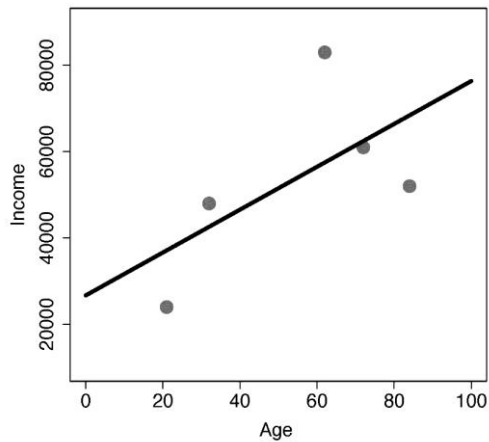
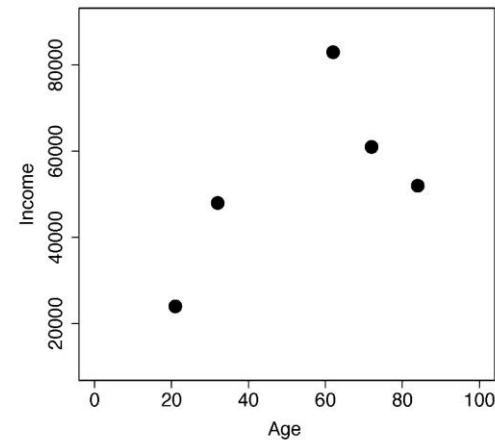
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- No free lunch!
 - Trade off
- What happens if we choose the wrong inductive bias:
 - Underfitting
 - Overfitting

Table 1.5: The age-income dataset.

ID	AGE	INCOME
1	21	24,000
2	32	48,000
3	62	83,000
4	72	61,000
5	84	52,000



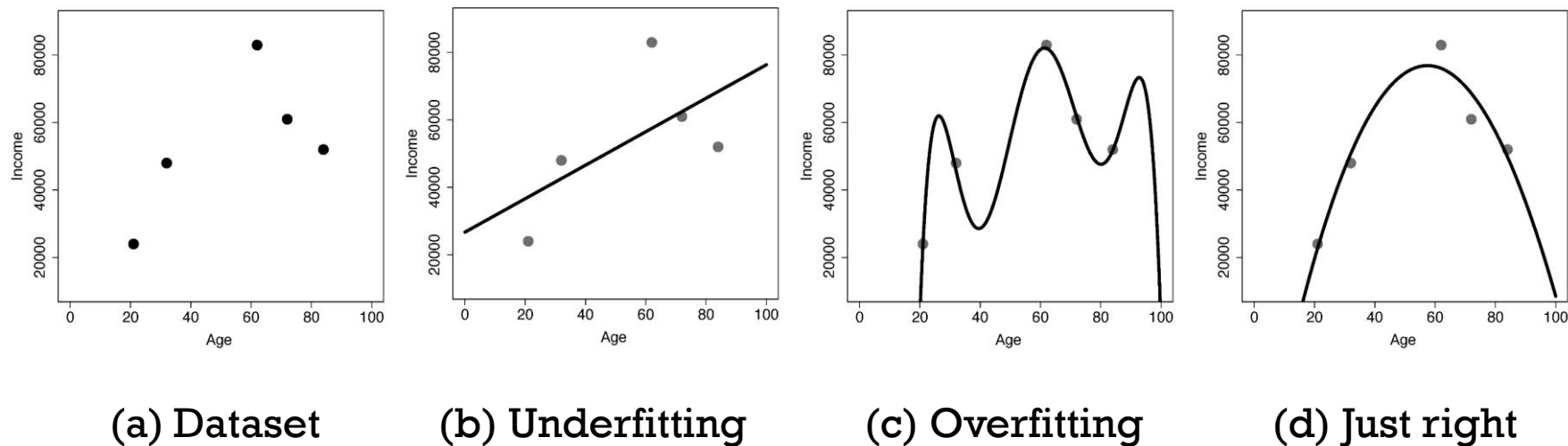


Figure 1.3: Striking a balance between overfitting and underfitting when trying to predict age from income.

- There are many different types of machine learning algorithms.
- In this course we will cover five families of machine learning algorithms:
 - Information based learning
 - Similarity based learning
 - Probability based learning
 - Error based learning
 - Deep Learning

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- One of the most commonly used processes for predictive data analytics projects is the **Cross Industry Standard Process for Data Mining (CRISP-DM)**

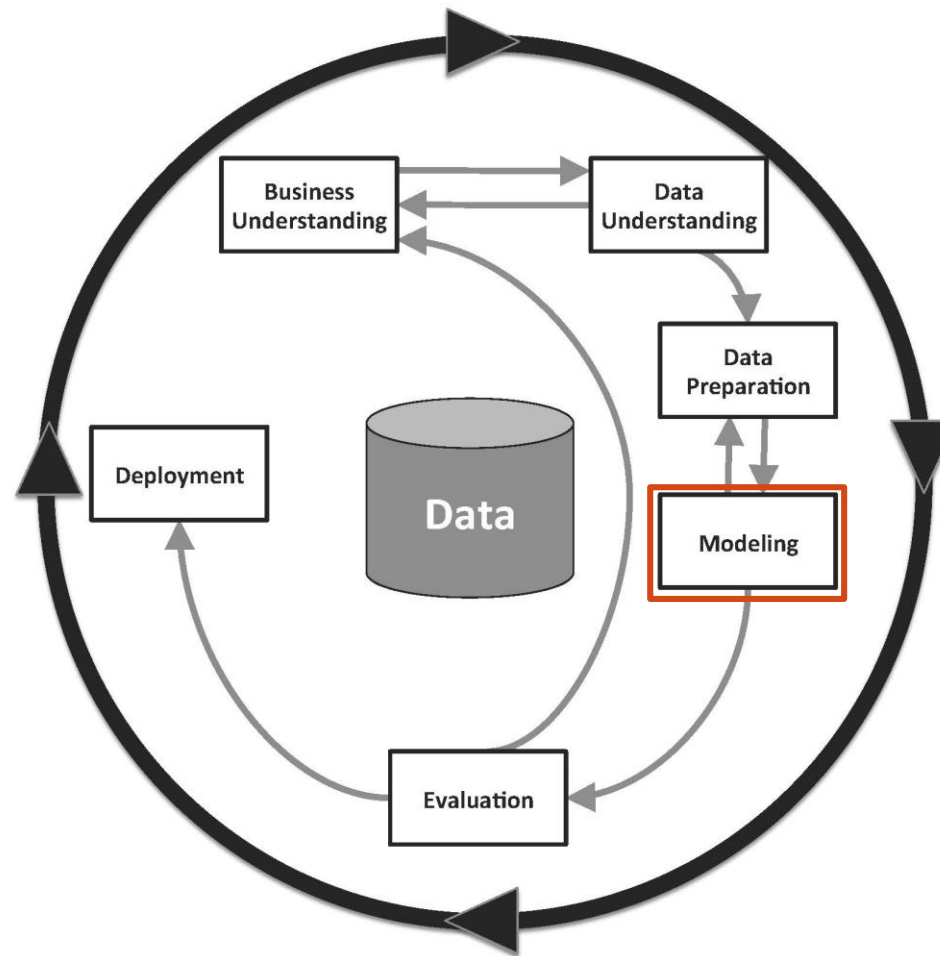


Figure 1.4: A diagram of the CRISP-DM process which shows the **six key phases** and indicates the important relationships between them.

■ Predictive Data Analytics Tools

■ Application-based solution

- IBM SPSS, Knime Analytics Platform, RapidMiner Studio, SAS Enterprise Miner, Weka...
- ✓ IBM and SAS are enterprise-wide solutions
- ✓ Knime, RapidMiner, and Weka are open-source, freely available solutions

■ Using a programming language

- Two of the most commonly used programming languages for predictive data analytics are **R** and **Python**
- ✓ Huge flexibility
- ✓ The newest advanced analytics techniques will become available

11:38   

Done IBM SlamTracker IBM

MEN'S SINGLES - QF Upcoming

Arthur Ashe Stadium

 B. Shelton [8]

 C. Alcaraz [2]

FIND TICKETS

Preview Live Summary

Likelihood to Win

 45% VS 55% 

B. Shelton C. Alcaraz

WHERE TO WATCH →

HEAD TO HEAD STATS

B. Shelton		C. Alcaraz	
0	CAREER H2H WINS	3	
United States of America	COUNTRY	Spain	
23	AGE	23	
9 October 2002	BIRTH DATE	5 May 2003	
Atlanta, GA, USA	BIRTH PLACE	El Palmar, Murcia, Spain	
6 ft. 4 in. (1.93 meters)	HEIGHT	6 ft. 0 in. (1.83 meters)	
Left Handed	PLAYS	Right Handed	
\$15,061,439 USD	PRIZE MONEY	\$64,997,598 USD	

FULL PROFILE FULL PROFILE

Profile data courtesy of ATP, WTA & ITF

PAST MATCH UPS

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- Machine Learning techniques **automatically learn** the relationship between a set of **descriptive features** and a **target feature** from a set of historical examples.
- Machine Learning is an ill-posed problem:
 - **generalize**
 - **inductive bias**
 - **underfitting**
 - **overfitting**
- Striking the **right balance** between model complexity and simplicity (between underfitting and overfitting) is the **hardest part** of machine learning.

HOMEWORK

- Reading: chapter 1
- Exercises: 1.9-4,9

1. What is **predictive data analytics**?

Predictive data analytics is a **subfield** of data analytics that **focuses on building models** that can **make predictions** based on insights extracted from historical data.

To build these models, we use machine learning algorithms to **extract patterns** from datasets.

2. What is supervised machine learning?

Supervised machine learning techniques automatically learn the relationship between a set of descriptive features and a target feature from a set of historical instances.

Supervised machine learning is a subfield of machine learning. Machine learning is defined as an automated process that extracts patterns from data.

3. Machine learning is often referred to as an **ill-posed problem**. What does this mean?

Machine learning algorithms essentially search through all the possible patterns that exist between a set of descriptive features and a target feature to **find the best model** that is consistent with the training data used.

It is possible to find multiple models that are **consistent** with a given training set (i.e., that agree with all training instances).

For this reason, **inductive machine learning** is referred to as an **ill-posed problem**, as there is typically not enough information in the training data to choose a single best model.

Inductive machine learning algorithms must somehow choose **one of the available models as the best**.

5. What is meant by the term **inductive bias**?

The inductive bias of a learning algorithm

- a. is a **set of assumptions** about what the true function we are trying to model looks like;
- b. defines the set of hypotheses that a learning algorithm **considers when it is learning**;
- c. guides the learning algorithm to **prefer one hypothesis** (i.e., the hypothesis that best fits with the assumptions) over others;
- d. is a **necessary prerequisite** for learning because inductive learning is an ill-posed problem.

6. How do machine learning algorithms deal with the fact that machine learning is an **ill-posed problem**?

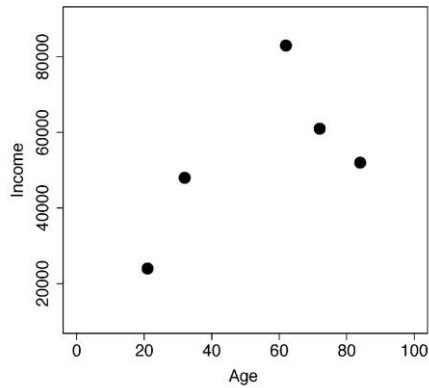
Because machine learning is ill-posed, we have to make some **extra assumptions** about what we think the real mapping from the descriptive features to the target feature should look like in order to find **a unique solution** with the data we have.

The set of assumptions we make so that learning is possible is called the **inductive bias** of the learning algorithm.

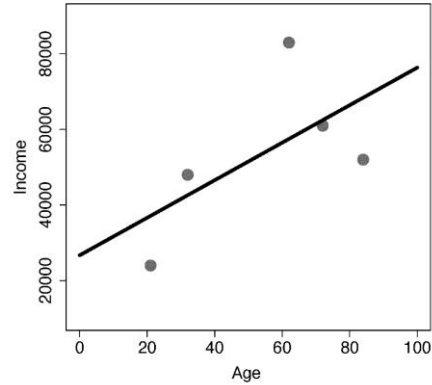
7. What can go wrong when an **inappropriate inductive bias** is used?

There are two possible negative outcomes:

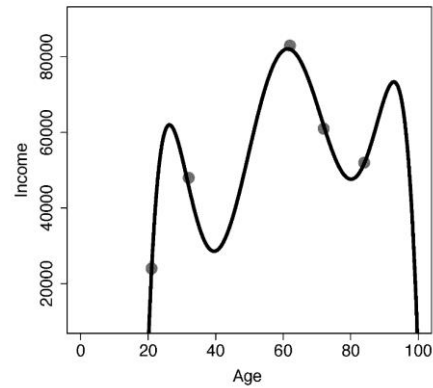
- If the inductive bias of a learning algorithm constrains the search to **consider only simple models**, we may exclude any models that successfully capture the relationship in the data from this search. In other words, the true model is unrealizable by the algorithm. In this case **underfitting** occurs.
- If the inductive bias of the learning algorithm allows the search to **consider overly complex models**, the algorithm may home in on irrelevant factors in the training set. In other words, the model will **overfit** the training data.



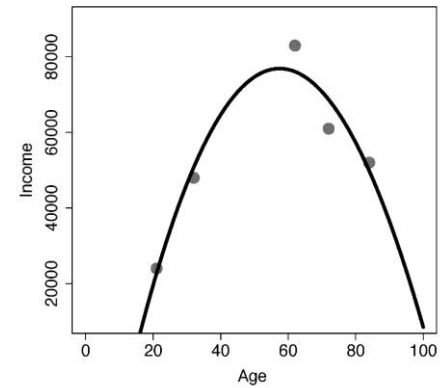
(a) Dataset



(b) Underfitting



(c) Overfitting



(d) Just right

8. It is often said that 80% of the work done on predictive data analytics projects is done in the Business Understanding, Data Understanding, and Data Preparation phases of CRISP-DM, and just 20% is spent on the Modeling, Evaluation, and Deployment phases. Why do you think this would be the case?

There is an old adage: “garbage in, garbage out.” No matter what machine learning algorithms are used or how they are tuned, it is not possible to build good models from bad data.

Because of this, it is crucial that sufficient time be spent gathering, analyzing, and preparing data before any modeling effort starts. This work is primarily done in the Data Understanding and Data Preparation phases of the CRISP-DM process.

The Business Understanding and Data Understanding phases of the CRISP-DM process also involve a huge amount of communication between the business domain experts and the predictive data analytics and machine learning experts in order to fully define the problem that will be solved. This communication is inevitably time consuming. It is, however, crucial so that the final models delivered are actually appropriate for the business.



Exploring the Frontiers of AI

Hod Lipson, director of Columbia Engineering's Creative Machines Lab, discusses the current state of AI and what's next. [MORE](#)

MAY 04 2023 | BY ROLAND WYN JONES | COLUMBIA BUSINESS MAGAZINE

CBS: Why is data so important to the development of AI?

Lipson: When you think about the economics of AI, certain things are a commodity and others are assets. Programming code, for example, is open source now and it's mostly free. Computing power is a commodity, too – a penny a core hour. And talent is ubiquitous. You have high school kids these days who can put together a system that would have earned them a PhD just a few years ago. People all over the planet (and AI itself) are learning quickly to create these systems.

But two things are not a commodity, and they're important to understand. One is data. The other is understanding what problems need to be solved. And this is where I think people with business backgrounds have an advantage. If you are in a particular market segment, you understand what the challenges are, and you understand what data are available to you. And you understand how data can help you solve a problem. Everything else you need is a commodity. You can put together a solution and lead your industry. This is what will differentiate the leaders from the followers in this world of AI.

CBS: How can generative AI help us achieve business goals or create new markets?

Lipson: The current wave of creative AI, or generative AI, is really fascinating. It's a different kind of intelligence. Up until recently, most of AI focused on decision-making. AI would ingest a lot of data and then make a decision. Is it a cat or a dog? Should I buy or sell? Should I turn left or turn right? Now, we are seeing a different kind of intelligence, which is creativity. You start from a goal, a seed, a very small thing, and then generate a lot of new things. We used to think computers can only make decisions and that creativity is uniquely human, but it turns out, creativity is exactly what AI is good at. It's actually very good at generating new ideas.

You can see that with software like ChatGPT, Stable Diffusion, or DALL-E. They can create not only poems, but music, scientific reports, engineering designs, molecules, or art. In class, I see students struggling to generate a new design for a robot, but AI can sketch out eight different designs in 25 seconds. It's amazing to see how creative AI can be. And this is important because a lot of our ability to innovate has to do with creativity. It's particularly difficult in areas where we don't have a lot of intuition. We are very good at designing chairs or bridges – the things we understand – but we are not as good at designing proteins or antennas, or nano materials – things that we don't have a lot of intuition about.